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THE ROLE OF URBAN SHRINKAGE DISCOURSE IN MULTILEVEL GOVERNANCE: EVIDENCE FROM THE RUSSIAN ARCTIC

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List of Abbreviations

ARK — Arkhangel'sk Oblast'

CHU — Chukotka Autonomous District

KAR — Karelia Republic

KHM — Khanty-Mansi Autonomous District

KOM — Komi Republic

KYA — Krasnoyarsk Kray

MUR — Murmansk Oblast'

NEN — Nenets Autonomous District

SAH — Sakha Republic (Yakutia)

YAN — Yamal-Nenets Autonomous District

Introduction

Problem statement

The Russian Arctic is a macro-region including 75 municipalities of 10 federal entities of the Russian Federation (ERDC, 2024). The current population of the Russian Arctic is approximately 2.41 million people (Rosstat, 2024), which is 36 per cent less than in 1989. According to the 1989 Soviet census, 3.78 million people lived in the contemporary borders of the Arctic Zone of the Russian Federation (Demoscope Weekly, n.a.). In general, the population decline in the Arctic for the last 35 years should be recognised as significant, averaging at –1.28 per cent per annum and rendering the Arctic a territory of shrinkage.

Despite the general trend towards population decline, the process of urban shrinkage in the Arctic exhibits a remarkable degree of heterogeneity (Zamyatina et al., 2023). While some cities are shrinking significantly, others are witnessing a considerable increase in their population size. Thus, the most shrinking cities in the Arctic — Igarka in the Krasnoyarsk Kray, and Inta in the Komi Republic — have seen a decline of approximately 4 per cent per annum during the 2010-2021 intercensal period. The most growing Arctic cities over the same period are cities of the Yamal-Nenets Autonomous Okrug. Gubkinskiy and Salekhard showed a population growth of more than 1 per cent per annum. The Arctic also consists of stagnating urban areas, where population loss is relatively slight. These include Anadyr' in the Chukotka Autonomous Okrug, with an average annual growth of 0.05 per cent, and Noril'sk in the Krasnoyarsk Kray, with an average decline of 0.05 per cent.

The Arctic region is significantly urbanised territory of Russia, with an urban population close to 90% (Zamyatina & Goncharov, 2020). The federal government is now paying close attention to urban development in extreme environments. Master plans have been designed for 16 key supporting settlements of the Arctic Zone of the Russian Federation. A substantial sum of 3 trillion rubles is earmarked for the implementation of these comprehensive urban development strategies (TASS, 2024a). Prioritised development of the Arctic Zone as well as development of the Russian Far East is linked to efforts aimed at 'preserving territories' in the context of their depopulation (TASS, 2024b).

The Arctic is being elevated to the status of a geostrategic region in the political discourse, which in turn is having an impact on governance practices. One of the most significant inquiries is the current status of the discourse surrounding urban shrinkage in the Arctic region — how the urban shrinkage problem is represented, who is responsible for the problem, and what measures are

contemplated to address the associated challenges in the policy formulation process. According to the concept of 'policy as discourse', problems are constructed by policies themselves as governmental responses to those issues (Bacchi, 2000). So, the matters of discourse formation are of paramount importance for comprehending policies. The large number of stakeholders associated with the Arctic, encompassing the national, and subnational governments, municipal administrations, and public and private entities investing in infrastructure development, underscores the critical nature of this issue.

Research problem

This bachelor's thesis contends that urban shrinkage is a multidimensional issue associated with the global challenge of depopulation. Population decline in the 21st century is a systematic and widespread phenomenon (Kinossian, 2018) that permeates social, political, and economic domains. It necessitates the involvement of actors at various territorial scales to devise more effective strategies for addressing this issue (Howlett et al., 2017).

The research on urban shrinkage has long been focused on examining administrative responses to population decline (Lee et al., 2023). Using qualitative methods in specific instances of shrinkage, scholars delve into why urban shrinkage is ignored or, conversely, is placed in the policy agenda (Coleman & Rowthorn, 2011; Wiechmann & Pallagst, 2012; Bernt et al., 2014; Hospers, 2014; Pallagst et al., 2017; Batunova & Gunko, 2018; Hartt, 2020).

To address this question, theoretical frameworks are employed, such as the concepts of 'policy streams' and 'policy windows' proposed by Bernt et al. (2014), which delineate specific conditions which turn an issue into a distinct policy. Additionally, the theory of urban regimes is also used as a framework for understanding policies in shrinking cities (Gunko et al., 2020).

Scholars also frequently allude to the institutional settings and the extent of local government autonomy, which dictate their capacity in terms of shrinkage addressing. The instances from South Korea, Germany, the Netherlands, Czechia, and China demonstrate how the cities' responses are heavily influenced by political decisions and budgetary allocations at higher levels, which in turn affect the development strategies of these cities (Joo & Seo, 2018; Bernt, 2019; Zhou et al., 2022; Bontje, 2024). Though 'shrinkage is a multiscalar process requiring multiscalar governance' (Bontje, 2024: 5), a significant amount of papers are still concentrated on the local policies (Bernt, 2019), precluding a more comprehensive vision of governance.

The third interpretation stems from the discourse framework in which policies are shaped (Haase et al., 2017). The concept of 'urban shrinkage' is rooted in the efforts of German planners to provide a neutral definition of this phenomenon to justify specific policies in depopulating eastern

cities (Bernt et al., 2016). Nevertheless, as Bontje (2024) notes, 'shrinkage' is still perceived negatively as something to be avoided or as a problem to be solved. In actuality, negative terms such as 'decline', 'crisis' or 'abandonment' are frequently employed when the process of shrinkage is discussed (Bernt et al., 2016: 442). Mallach (2017) shows that urban shrinkage in the United States is being portrayed as an opportunity for future growth.

As some authors note (Rink et al., 2014; Haase et al., 2016; Batunova & Gunko, 2018; Chigareva, 2022), there has been a long-standing avoidance of urban shrinkage in Russia and other former socialist countries. In the term paper of Lettieva (2024), it was demonstrated that authorities of the Komi Republic — one of the Arctic regions of Russia — have begun to employ the concept of 'controlled shrinkage' to excuse forced measures of increasing budget expenditure efficiency.

The problem with accepting or ignoring shrinkage may be related to policy formulation as one of the elements of the policy process. At this stage, problems and solutions are essentially created (represented), policy tools are selected, and responsible subjects are identified. However, studies focusing on policy formulation have not yet been presented. The study aims to trace the characteristics of discourse regarding urban shrinkage and its impact on multilevel policy in the Arctic.

Research question: What are the characteristics of discourse on urban shrinkage, which affect multilevel policy in the Arctic?

Research object: discourse on urban shrinkage in the Russian Arctic.

Research subject: the role of urban shrinkage discourse in multilevel governance in the Russian Arctic.

Research aim: to identify the role of urban shrinkage discourse in multilevel governance.

Research tasks

1. To ground the theoretical framework of the research and systematise the existing body of knowledge on governance under population loss and policymaking process;
2. To present population dynamics at different territorial scales based on official statistics;
3. To contextualise public policy in the Arctic Zone of the Russian Federation;
4. To identify the main trends of urban shrinkage problem representation in the Arctic through qualitative content analysis of discourse in the news media publications;

5. To identify discourses on urban shrinkage through the analysis of strategic planning documents from different territorial levels;
6. To compare the results of media analysis and analysis of documents and determine the role of discourse on urban shrinkage in multilevel governance.

Hypothesis

The public discourse on urban shrinkage in the Russian Arctic influences policy formulation by making the problem representation unclear and the policy response inconsistent.

Chapter 1. Literature Review

1.1 Urban Shrinkage as a Challenge for Multilevel Urban Governance

The phenomenon of urban shrinkage represents a relatively novel field of research, with a surge in attention towards the issue of shrinking cities emerging at the beginning of the 21st century (Turok & Mykhnenko, 2007; Hollander et al., 2009; Pallagst, 2010). Before this, the problem of urban population decline was addressed in the theoretical model of cyclic urban development, exemplified by 'A Study of Growth and Decline' by van den Berg et al. (1982). Within this model, 'urbanisation' and 'desurbanisation' were distinguished as natural phases of urban development.

Another perspective views urban shrinkage as a consequence of global processes and the increasing significance of metropolises (Martinez-Fernandez et al., 2012). According to Brenner (2004), globalisation towards the end of the 20th century fostered the hierarchical arrangement of cities in European countries. Thus, the global urban system can be represented as a pyramid, with the most developed global centres at the top and small peripheral cities forming the foundation (Mykhnenko & Wolff, 2019).

Migration due to the economic decline in hinterlands and uncompetitive territories is often cited as a reason for urban population decline (Wiechmann & Bontje, 2015). Indeed, deindustrialisation is the most common cause of urban shrinkage in Western countries (Döringer et al., 2020). Examples include the old industrial cities of the USA ('Rust Belt') and Great Britain (Liverpool, Manchester) (Hollander et al., 2009; Rink et al., 2012). Another instance is the industrial socialist cities of Eastern Europe, which underwent significant structural transformations (Haase et al., 2016). Additionally, one of the primary reasons for urban shrinkage is the tendency towards population decline caused by a decrease in the birth rate below the level of population reproduction and population ageing (Coleman & Rowthorn, 2011; Wiechmann & Bontje, 2015), which leads to population decline in developed cities, as exemplified by Japan's ageing cities (Döringer et al., 2020).

There is a lack of universally accepted understanding of the urban shrinkage process and a fixed set of attributes that characterise a shrinking city (Bernt et al., 2014). The quantitative evaluation of shrinkage remains ambiguous and in recent studies, it is often abandoned. The mere fact of population decline in cities is considered sufficient, particularly in qualitative research. A widely accepted definition could be formulated as follows: urban shrinkage is 'population decrease within a defined urban territory' (Rieniets, 2009: 236).

The main consequences of urban shrinkage are the housing crisis, leading to an accumulation of vacant properties, rising unemployment rates, a shrinking tax base, and falling city budget revenues, which can result in underinvestment in urban infrastructure. Additionally, cities may face stigmatisation (Rieniets, 2009; Döringer et al., 2020). It is crucial to note that both the causes and consequences of urban shrinkage are diverse and contingent upon regional specificities.

In a recent meta-analysis, Lee et al. (2023) highlighted that research on urban shrinkage in the fields of policy and governance has gained significant prominence. A substantial empirical foundation for research on governance under population loss was provided by the case of East Germany: between 2002 and 2016 the federal program 'Stadtumbau Ost' (cities reconstruction of the East) was implemented in response to the economic and housing crisis in the shrinking Leipzig and other post-socialist cities of Germany (Rink et al., 2012; Bernt, 2019). Since then, many studies have emerged clarifying the contexts and practices of planning or governing in shrinking cities across North America, Western and Eastern Europe, and Asia (Hollander et al., 2009; Wiechmann & Pallagst, 2012; Bernt et al., 2014; Hospers, 2014; Haase et al., 2016; Stryjakiewicz & Jaroszewska, 2016; Pallagst et al., 2017; Batunova & Gunko, 2018; Joo & Seo, 2018; Hartt & Hackworth, 2020; Zhou et al., 2022; Liu et al., 2022; Bontje, 2024).

Policy responses to urban shrinkage are distinguished into four categories: (1) trivialising shrinkage, (2) countering shrinkage, (3) accepting shrinkage, and (4) utilising shrinkage (Hospers, 2014). Nevertheless, Eraydin and Özatağan (2021) argue that these categories represent stages of adaptation rather than distinct strategies. Researchers conceptualise urban shrinkage adaptation as a three-phase process: (1) 'recovery' as an initial response to restore growth, (2) 'adaptation' as a policy for incremental economic reorientation, and (3) 'transformation' akin to a 'smart shrinkage' characterised by a comprehensive economic diversification and a decrease in the size of the city.

The majority of studies on urban shrinkage tend to concentrate on case studies, as the contextual factors of urban development, such as geographical and political circumstances, significantly influence policymaking in cities (Bernt et al., 2014; Wiechmann & Bontje, 2015; Haase et al., 2016). Bernt (2019) emphasised that urban policy ought to be viewed within the broader context of multiple territorial scales, particularly given that population decline and urban shrinkage constitute a global trend (Hospers, 2014; Mykhnenko, 2023). As Bontje (2024) notes, 'shrinkage is a multiscalar process requiring multiscalar governance' (Bontje, 2024: 5).

Nonetheless, urban policies in shrinking cities are seldom explored in a multilevel perspective. Multilevel governance can be defined as 'an instance of policymaking in which government(s) engage with a variety of non-governmental actors, organised at different territorial scales, in a process of decision-making that aims to collaboratively produce some sort of public

good' (Alcantara et al., 2016: 7). Notable examples include Joo & Seo's (2018) study, which delves into the dual urban shrinkage policies in South Korea; Bernt's (2019) examination of how multilevel policy adaptation to shrinkage in Leipzig (East Germany); and Zhou et al.'s (2022) analysis of multilevel interactions in shrinking cities across China, the Czech Republic, and the United States. Additionally, Bontje's research (2024) examines the outcomes of a multilevel program aimed at coping with urban shrinkage in the Netherlands.

The studies on urban governance under the shrinkage process in Russia are relatively new and limited in number (for instance, Batunova & Gunko, 2018; Gunko et al., 2020; Zamyatina et al., 2020b; Chigareva, 2022; Zamyatina et al., 2023). In many cases, researchers conclude that the urban shrinkage problem remains largely unaddressed in Russia. Lee et al. (2023) suggest that there is a lack of understanding of strategies to tackle urban shrinkage challenges in *Russia*, Australia, Iran, South Korea, and Canada. In the following section, I delve into the literature on urban shrinkage within the Russian Arctic.

1.2 Urban Shrinkage in the Russian Arctic

The development of the Russian Arctic has been highly uneven. Its mastery began long before the Soviet intervention, however, significant urban growth took place during the USSR period. According to the Arctic shrinkage expert, Nadezhda Zamyatina, key milestones in Arctic urbanisation can be traced to waves of resource exploitation. The first phase, spanning from the 1920s to 1940s, saw the onset of Raw Material Colonisation characterised by complex urban development. The second phase, nearly between the 1950s and 1980s, marked the advent of single-industry colonisation in the Far North.

Cities that emerged during the first phase of Raw Materials colonisation often lost their diversity and further their main resource. Examples include cities such as Vorkuta, Inta, Kirovsk, and Igarka, which experienced a significant population decline of more than 40 per cent. In contrast, cities that emerged during the second phase specialised in extracting hydrocarbons. Some of these cities have seen population growth due to the demand for energy resources and skilled labour.

Nonetheless, during the 1990s, there was a significant depopulation of cities and work settlements in the Far North, which was attributed to the closure of city-forming enterprises, the transition of cities to special-shift work status, and the abandonment of entire settlements (Zamyatina et al., 2023). The contemporary Russian Arctic is primarily a territory of shrinkage. The prevailing narrative in explaining the evolution of Arctic cities through external forces fails to

acknowledge their resilience to challenges (Zamyatina et al., 2020a). Consequently, it is not merely geographical conditions that matter, but also the governance strategies employed in the Arctic.

It is worth noting that the Arctic is one of the most urbanised areas in Russia, with a typical resident being an urban one (Zamyatina, 2017; Zamyatina & Goncharov, 2020). According to Zamyatina (2017), 'Arctic urbanisation should be viewed as a frontier, a boundary of urbanistic (Zamyatina, 2017: 23). This frontier exhibits a constant fluctuation: the population in cities experiences dramatic declines followed by sharp increases, while the economic conditions for development, particularly in the raw materials industries, undergo constant change. While some cities continue to face severe housing shortages, others are confronted with significant problems with infrastructure overcapacity (Zamyatina et al., 2020b).

The key aspect in the Arctic cities' development is their remoteness, rendering the Russian Arctic a remote region. Due to remoteness, Arctic cities lack numerous advantages associated with 'normal' urban development, such as agglomeration effects, synergistic interactions, extensive networking, and knowledge exchange (Zamyatina et al., 2017). As Zamyatina (2017) suggests, three key elements are essential for the sustainable development of Arctic cities: (1) mobility of the population, (2) adaptability, and (3) creativity. In the Arctic, more than anywhere else, cities need to be «smart» to ensure their survival and viability.

Indeed, adaptability is an essential prerequisite for the survival of urban systems in the face of unstable and challenging socio-economic and climatic conditions in the Arctic. The Vorkuta city in the Komi Republic stands out as a striking and well-known example of urban shrinkage adaptation in Russia (Shiklomanov et al., 2020; Gunko et al., 2021). In 30 years, the Vorkuta agglomeration has witnessed a loss of nearly 70% of its population, resulting in widespread desolation, particularly on the periphery. The municipality has been burdened with the maintenance costs of the semi-vacant housing stock of Soviet industrial construction. In response, the municipality adopted an independent strategy known as 'controlled shrinkage', aimed at a compact urban core created through the relocation of residents from the outlying areas and the preservation of vacant houses.

The Komi Republic serves as a crucial case study for examining the urban shrinkage policy adaptation. Remote industrial cities of the Komi Republic, such as Inta and Vuktyl, have been forced to reshape their strategies to address shrinkage (Lettieva & Shtinov, 2023). Officials from regional ministries deem 'controlled shrinkage' to be the dominant trajectory for the region's development (Lettieva, 2024). Furthermore, the Komi Republic has proposed amendments to the federal legislation on housing to facilitate population densification in the Arctic.

It is noteworthy that issues related to urban shrinkage policies have received scant attention in Russia. Urban shrinkage has been examined more extensively within the domain of geographical inquiry rather than being a subject of policy analysis. The distinctive characteristics of multilevel governance in Russia play a pivotal role, as power and financial resources are concentrated at the federal level while spending responsibilities are dispersed to the regional level (Yushkov et al., 2017). This approach renders urban policy heavily dependent on regional and federal ones.

In the context of the Arctic, policy actors, especially federal, play a huge role for urban development. However, as foreign samples show, a balance of power may be differnet: for example, in the Arctic city of Kiruna, Sweden, the ideology of expansion was promoted by the mining company, Luossavaara-Kiirunavaara Aktiebolag (Nillson, 2010). The decision to shrink the city was also made by the company. In Canada, the urban shrinkage policy in the Arctic is much more localised and adaptive. Cities in the Klondike region, Yukon (for instance, Dawson) exist in conditions of cyclical demand for resources with rises and declines (Vaguet et al., 2021).

This bachelor thesis explores urban shrinkage from a novel perspective, considering policy formulation at various levels of governance. To achieve this, it is important to incorporate the literature on the policymaking process in the field of policy studies. In policy analysis, urban shrinkage may be presented as a 'wicked problem' — a messy, fuzzy, complex multi-causal problem that requires intersectoral interventions (Bacchi & Goodwin, 2016).

The way policy is formulated affects the decision-making process and, consequently, policy results. The relationship between discourse and policy in the context of urban shrinkage has been already highlighted by Haase (2017) and Mallach (2017). Discourses represent certain issues, and through that representation provide the frame for policy responses. It is posited that a discursive view on policy provides insights into how the urban shrinkage issue is represented and how policy tools are chosen to address the challenge of depopulation.

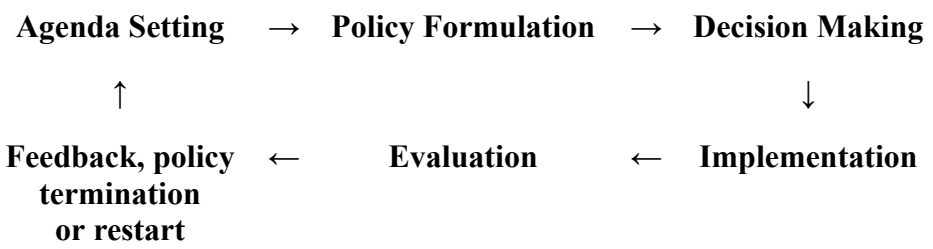
1.3 Public Policymaking and Policy Design Theory

Public policy can be defined as 'A statement by the government — at whatever level, in whatever form — of what it intends to do about a public problem' (Birkland, 2019: 6). The absence of statements can also be an implicit expression of a policy aimed at not doing something (Ibid). As a rule, statements are presented in the form of laws, regulations, decisions, and other forms.

The most comprehensive attempt to model the policymaking process was presented in the 1950s in Lasswell's work, which has since been refined and modified. Jann & Wegrich (2017) propose a widely accepted framework for policy development — policy cycle, comprising six

stages: (1) agenda setting, (2) policy formulation, (3) decision-making, (4) implementation, (5) evaluation, (6) feedback, and policy termination or restart (Figure 1). In essence, this model presents policymaking as a linear process, which has been criticised for its oversimplification. Political processes rarely follow a straightforward sequence, and it is often difficult to delineate the various stages of policy development (Hill & Varone, 2021).

Figure 1. Simplified model of the policymaking process



Source: Jann & Wegrich (2017)

Agenda Setting. Jann & Wegrich (2017) suppose that the first three components of the policymaking process have the greatest impact on policy itself. As regards agenda setting, it is conceptualised as 'the process by which problems and alternative solutions gain or lose public and elite attention' (Birkland, 2017: 63). The classical definition is attributed to Kingdon (1984): Agenda is 'A list of subjects or problems to which government officials, and people outside the government closely associated with those officials, are paying serious attention at any given time'. The salient feature of this process is limited system capacity, which precludes the simultaneous consideration of all issues, thereby necessitating social groups to vie for a place on the policy agenda.

Cobb & Ross (1997) suggest that it is not the problems themselves that are in competition but rather how they are interpreted. Thus, in the agenda-setting stage, the creation of issues occurs, which involves defining policy problems that are not solely determined by objective circumstances. The American scholar Schattschneider noted in the 1960s that a social group most effectively articulates a problem and proposes solutions, thereby prevailing in policy debates (Mair, 1997; Birkland, 2017). Moreover, influential policy actors can effectively block specific issues from being addressed.

Policy Formulation — Policy Designing. The next stage is of particular significance. Policy formulation enables scholars to broaden the discussion on the policy process. At this stage, an action plan emerges for addressing the issue: what policy alternatives exist, what policy tools are at the disposal, and how should the solutions respond to the challenges (Sidney, 2017). Policy formulation could be described as a process examining 'various policy options or alternatives available for addressing a problem' (Turnpenny et al., 2015: 4).

Bacchi (1999), and Hill & Varone (2021) propose two primary frameworks for policymaking: the rational comprehensive model (technical rationalists) and the politically rational model (incrementalists) (Table 1). These two approaches can be meaningfully integrated into the policy formulation process, as they seek to address fundamental questions such as: what is the essence of the policy, how and by whom the issue is defined, what should be the objectives and instruments of the policy, how a set of values is formed, and what is the solution?

Table 1. Policymaking models

	The rational comprehensive models *(Technical rationalists)*	Politically rational models *(Incrementalists)*
Epistemology	Positivist: there is a real-world accessible to objective description and analysis.	Transition to post-positivist: reality is framed by certain references, but facts are waiting to be discovered.
Approaches to policy	Problem identification	Problem definition
Policy Analyst	Technical expert	Conflict manager
Type of power	Centralised	Pluralistic
Policy objectives (ends) and instruments (means)	'The ideal rational model postulates the prior specification of ends (perhaps we may call this "agenda setting") and the identification of means of reaching these ends' (Hill & Varone, 2021: 146)	Decision makers adjust objectives to available means instead of striving for a fixed set of objectives.
Values dilemma	A policymaker must pursue a consistent set of values. A stable set of values and goals is clarified and specified at the first stage of the formulation process.	The practical impossibility of a stable set of values, as negotiations and compromise between complementary and contradictory values and goals continue throughout the decision-making process.
Problem solution	There is the best collective solution that meets the public interest, which can be rationally and analytically determined.	For most political decisions, there can never be a definitive solution, and the problem will remain unresolved no matter what policy is implemented.
Policy essence	1) Decision making, finding solutions to problems. 2) Problem identification/definition and problem solution are separate elements of policy formulation	

Source: author's work based on Bacchi (1999), Hill & Varone (2021)

Theoretical models are grounded in diverse conceptualisations of dichotomies such as 'problems — solutions', 'policy objectives — policy means', and 'values — facts'. Technical rationalists maintain that problems exist objectively, whereas political rationalists contend that

problems are primarily determined. Both perspectives acknowledge a degree of separation between problems and solutions, and both view the essence of policy in problem-solving.

Regarding policy objectives and means, technical rationalists advocate for their clear definition by experts during the policy formulation stage, whereas political rationalists emphasise the need for continuous adjustment of aims and instruments due to ongoing social conflicts in the policymaking process.

Value considerations are largely trivialised in the technical rationalist view, with participants in the policymaking process expected to adhere to a predetermined set of values. Political rationalists acknowledge the paramount importance of values and maintain that stakeholders ought to perpetually strive for a harmonious equilibrium of power distribution and to enhance policymaking.

Both approaches are limited as they assume certain social constructs, such as the characteristics of target groups, values, motivations of actors, and abstract policy objectives, to be self-evident. This assumption, in turn, results in the promotion of perspectives from the dominant social groups (Fischer, 2003). It is crucial to acknowledge that not only subjective elements but also the policy issues themselves are not inherently clear or neutral. They are socially constructed and may change through time in response to societal values (Turnpenny et al., 2015: 4).

The branch of policy studies that explores policy formulation is known as policy design theory (Sidney, 2017). According to Linder & Peters (1987), difficulties in policy formulation are one of the primary causes of policy failure, highlighting the importance of effective policy design. Policy design involves the systematic collection and application of knowledge about the effects of policy tools on policy aims for policy development and implementation (Howlett et al., 2015: 291). The supreme policy formulation is determined by a supreme set of policy tools (Howlett & Mukherjee, 2018). In practice, the goal of this theory is to devise improved strategies for addressing issues and achieving optimal outcomes within limited administrative resources (Peters et al., 2018).

Policy design theory allows for the integration of social construction processes into the process of policymaking (Schneider & Sidney, 2009). 'Social Constructions and Policy Design' is a distinct area of policy process research (Nowlin, 2011), which focuses on the role of social constructions in shaping policy. Social constructions refer to a set of attributes that have been historically formed concerning specific objects, subjects, and issues. Social constructionism posits that multiple interpretations of reality contribute to the social reality creation, making it crucial to consider these interpretations in contemporary policy studies (Ingram et al., 2019).

Why is social construction significant in policy research? According to Birkland (2017), the way a problem is defined significantly influences the solution selection. Moreover, the

interpretation of a problem is closely linked to the prevailing social, political, and ideological frameworks at the time of policymaking. This leads to a postmodern and post-positivist perspective on policymaking that emphasises the examination of policy's role in social construction.

1.4 Social Construction of Problems and Policy as Discourse Concept

The concept of social construction is far beyond the traditional approaches to policy analysis and policy cycle, which are epistemologically based on the existence of knowable objective reality (these approaches are often critiqued by postmodernists). In essence, social construction rejects positivism and empiricism and excludes the objective, unbiased observation of social reality (Burr, 2015: 2). Bacchi (1999) argues that policy formulation should be viewed from a constructivist perspective, emphasising the discursive creation of problems. Fischer (2003) goes further, suggesting that public policy is not merely expressed through words but is constructed through language. He posits that social institutions cannot exist independently of discourse, as widely accepted intersubjective meanings become embedded within societal institutions and practices (Fischer, 2003: 44).

Among the proponents of the discursive approach to policy analysis, authors such as Ball (1993, 2015), Fischer (2003), Yanow (1995, 2007), and Bacchi (1999, 2000, 2009, 2016) can be highlighted. These scholars draw upon Foucauldian ontology, which establishes a close relationship between knowledge, discourse, and power. In the realm of classical perspectives on the policy as discourse, the theory of hegemonic discourse by Laclau and Mouffe (Leipold & Winkel, 2017) warrants special attention. According to the approach proposed by Foucault, discourses do not merely represent reality but actively contribute to its construction (McHoul & Grace, 2015: 35). Consequently, the subject or object of policy cannot be adequately understood without considering its discursive constructions.

Discourse can be understood as 'practices which form the objects of which they speak' (Ball, 2015). Moreover, discourse entails a 'set of meanings, metaphors, representations, images, stories, statements and so on that in some way together produce a particular version of events' (Burr, 2015: 74).

The social meanings themselves often fall within the interpretive approaches to policy (Yanow, 1995; 2007). As she states, 'Meanings are never treated independently of the contexts of people doing and saying the things that created those meanings' (Yanow, 1995: 112). Yanow provides a crucial starting point for policy analysis, arguing that there are no objective facts that exist outside the frameworks of perception and language. Therefore, Congressional hearings cannot

be simply viewed as a collection of isolated facts. Instead, these hearings offer a platform for shaping, refining, and transforming interpretations of political matters. Furthermore, social meanings (motives, intentions, objectives, and values) are always potentially open to reconstruction and change (Fischer 2003: 49).

Ball (1993) pioneered two post-structuralist concepts: 'policy as text' and 'policy as discourse'. In the former, policy is conceptualised as 'textual interventions into practice' (Ball, 1993: 12), which implies that it narrows or alters the range of options available for decision-making. In the latter, 'policy as discourse' not only creates multiple meanings but also defines the usage of language, consequently limiting the potential for thought. This aligns with Foucault's perspective, where discourse represents a specific body or form of knowledge that restricts what can be thought, written, or spoken about a particular object or practice (Bacchi & Goodwin, 2016: 35). Thus, discourse and power are intricately linked.

Continuing to elaborate on the concept of Ball, Bacchi (1999, 2000, 2009) highlights that the challenges in policy fostering do not primarily arise from conscious counteraction to reforms, but rather from the results of how policy problems are represented. She posits that the impact of representation can constrain policies and ongoing transformations (Bacchi, 2000: 46). Her approach 'What's the problem represented to be?' is predicated on the social construction of policy issues, rejecting the identification or discovery of social problems, and emphasising that policy is a discourse through which governments both create problems and solutions (Bacchi, 2009).

Significantly, policy shapes the world by framing social 'problems' and government 'solutions' and the construction of concepts, categories, distinctions and subject positions (Goodwin, 2011: 170). Bacchi's approach seeks to answer fundamental questions such as: what is considered a 'problem'? Who or what is perceived as the source of this 'problem'? (Bacchi, 2009; Van Hulst et al., 2024).

This perspective allows us to view policy as a set of discursive practices, necessitating an analysis of policy texts (Goodwin, 2011). As the author pointed out, policymaking is a far more complicated process than merely responding to objective data, such as statistics on population decline. This thesis demonstrates that the urban shrinkage issue is not trivialised in Russia. The point is not simply that decision-makers fail to pay attention to statistical data or cope with the consequences, but rather how they formulate the problem and address it. The main idea is that discourse on urban shrinkage represents certain issues, and through that representation provides the frame for policy responses (Haase et al., 2017). Thus, It is of particular interest to examine how the issue of urban shrinkage is represented and how it shapes policy in Russia, specifically in its Arctic part, which serves as a testing ground for innovative governance.

Chapter 2. Context: Policy Change in the Arctic

Recently, the Russian Government has been paying great attention to the Arctic. In 2014, under a decree issued by the President of the Russian Federation, the territorial boundaries of the Arctic Zone were delineated (President of the Russian Federation, 2014). Subsequently, from 2017 through 2020, these boundaries were further expanded through a series of presidential decrees (President of the Russian Federation, 2017; 2019). In 2019, the Ministry for the Development of the Far East was renamed the Ministry for the Development of the Far East and the Arctic. This implies that the Arctic is governed more centralised with the aim of fostering growth (Libman & Yakivlev, 2021).

The cornerstone of the Arctic's development lies in megaprojects undertaken through public-private partnerships in the domains of transportation and oil and gas production. This was stated in previous versions of the current documents: Strategy for Development of the Arctic Zone of the Russian Federation and Providing National Security for 2020 (President of the Russian Federation, 2013), Russian Federation state programme of Socio-Economic Development of the Arctic Zone for 2020 (Government of the Russian Federation, 2014b).

In 2020, the Fundamentals of the State Policy of the Russian Federation in the Arctic for 2035 were adopted, as well as the Strategy for the Development of the Arctic Zone of the Russian Federation and Ensuring National Security for 2035 (hereinafter referred to as the Arctic Strategy) (President of the Russian Federation, 2020a; 2020b). In 2021, the Government issued the State Program of the Russian Federation 'Socio-Economic Development of the Arctic Zone of the Russian Federation' (Government of the Russian Federation, 2021), which was designed to ensure the implementation of the Arctic Strategy objectives. It also serves as a blueprint for regional-level strategic planning.

As Kinossian (2017) notes, in the 2010s Russia's Arctic policy shifted from a phase of withdrawal to one of renewed participation through megaprojects in energy and transportation. This shift and subsequent refocus of attention are linked to the discovery of vast potential for national economic growth, which great Arctic resources possess. The resurgence of collaboration between the Government and the Arctic is driven by a desire to regain lost influence, secure access to its resources, and reverse population decline. Following Kinossian (2017), this process can be described as a 're-colonisation' of the Arctic. Thus, the issue of population dynamics in the Arctic holds significant importance.

Nonetheless, the multilevel policy design in the Arctic remains somewhat obscure. The resolution of issues in the Arctic has been declared as a collaborative endeavour, delineated by the boundaries of regional and municipal jurisdictions. However, Arctic strategic objectives typically

extend beyond many other policy actors and sectors. Consequently, the perception of Arctic development remains different. Kinosian (2017) underscores the fact that while the subnational level does not serve as an unequivocal conduit for federal policy, it also does not operate as an independent actor. According to him, Arctic regions are 'produced via multiscalar processes of policymaking, institutionalisation, and discursive practices' (Kinossian, 2017: 221).

Indeed, the Arctic policy is in the period of its formation and institutional consolidation — a creation of thematic committees in legislative bodies, commissions, designing of a national project, etc. Subsequently, it is still a large discursive field in which ideas are intertwined. The ideal image of the Arctic was announced by President Putin at the plenary session of the VI International Arctic Forum in Murmansk (Kremlin.ru, 2025). It follows from his words that the future Arctic is characterised by high rates of extraction and deep processing of raw materials, an ultra-efficient transport system, as well as comfortable growing cities.

Thus, there are national-level objectives linked to the development of Arctic and maritime routes and the expansion of the national economy that are endorsed by federal and regional governments. Conversely, there is the depopulation issue, which presents a more pressing challenge at the subnational level. This highlights the potential ideological ambiguity surrounding the approach to shrinkage addressing: how should the problem be represented, and what means should be implemented? Therefore, it is crucial to consider a multilevel discourse in the Arctic.

The ideological dimension of the paradigm shift in Arctic policy cannot be overstated. As Wilson (2000) observes, from an ideological and cultural perspective, the paradigm shifts are accompanied by changes in the way policy problems are defined and the types of policy solutions presented. This approach implies that paradigm shifts also entail changes in officials' perspectives.

The discursive approach emphasises the importance of examining how policy actors perceive, comprehend, and articulate policy concerns, as well as their interpretations of ideas and their role in the political discourse (Swinkels, 2020: 285). Political actors actively construct ideas about problems and solutions through discourse (ibid.), and they also shape themselves through discourse.

Thus, in the face of the lack of a comprehensive response to the urban population decline in the Arctic, it becomes imperative to examine the discourse on urban shrinkage, which is shaped by political actors operating at both the national and subnational levels.

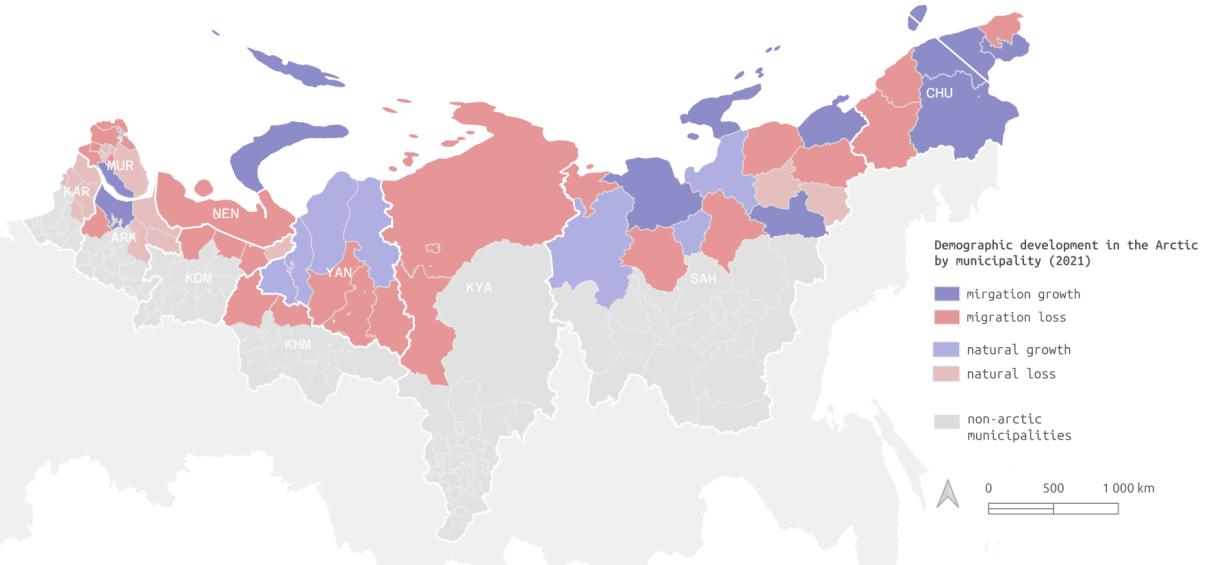
Chapter 3. Data and Method

3.1 Quantitative Assessment of Urban Shrinkage in the Russian Arctic

The extent of population decline in the Russian Arctic is noteworthy. Over a period of 32 years, the population has decreased by 29.8 per cent (Demoscope Weekly, n.a.; Rosstat, 2021). For the purposes of the study, the demographic situation within the municipalities of 10 Arctic regions was examined and the population dynamics between 1989 and 2021, including a specific focus on the intercensal period from 2010 to 2021, was analysed.

Figure 2 presents an overview of the current demographic landscape in the Arctic as of 2022, taking into account the results of the 2021 Russia Population census. The governmental database of municipal indicators provides data on both natural and migratory population growth. By analysing the dynamics within each of the 75 municipalities, the prevailing trend in demographic changes was identified.

Figure 2. Demographic development in the Russian Arctic, 2021



Source: author's work based on governmental database of municipal indicators[1]

The pie chart in Figure 3 illustrates the demographic structure in the Arctic. It is evident that the decline is primarily driven by out-migration losses, and there is also a significant natural decline caused by decreased birth rates and high mortality rates.

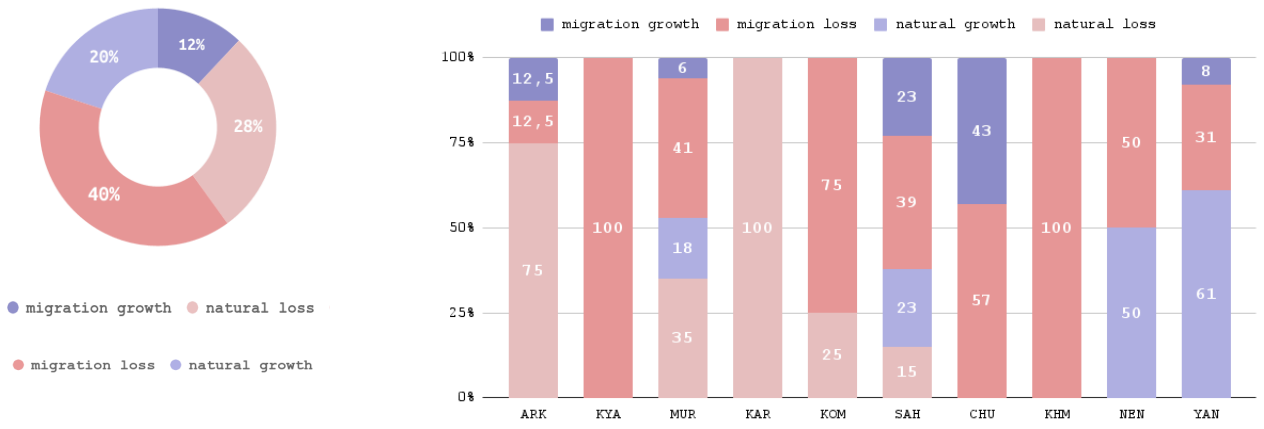
As depicted in the bar chart in Figure 3, migration losses account for the majority of the decline in 7 out of 10 Arctic regions. In the Krasnoyarsk Kray (KYA) and the Khanty-Mansi

[1] Governmantal database of municipal indicators is available. URL: https://rosstat.gov.ru/dbscripts/munst/

Autonomous Okrug (KHM), the proportion of migration losses is 100 per cent. In the Komi Republic (KOM) it stands at 75 per cent. The group experiencing a decline due to natural loss includes the Arkhangel'sk Oblast' (ARK) and the Karelia Republic (KAR). Additionally, in these two regions, the population decline has seen a notable increase compared to the rest of the regions during the period from 2010 to 2021, as illustrated in Figure 4.

A relatively high rate of natural growth is observed in the key oil and gas-producing regions, such as the Nenets Autonomous Okrug (NEN) and the Yamal-Nenets Autonomous Okrug (YAN). The Murmansk Oblast' (MUR) has traditionally been a region characterised by high levels of out-migration. However, in recent years, a trend of natural decline has started to predominate in many of its municipalities. The Chukotka Autonomous Okrug (CHU) has stood out as the region with the highest rate of population decline. Nevertheless, there has been a migratory increase within this region. The demographic situation in CHU has also been mitigated in the period from 2010 to 2021 (Figure 4). Sakha Republic (Yakutia) (SAH) is a region that exhibits all demographic trends.

Figure 3. Demographic trends in Arctic municipalities at whole and by region, %



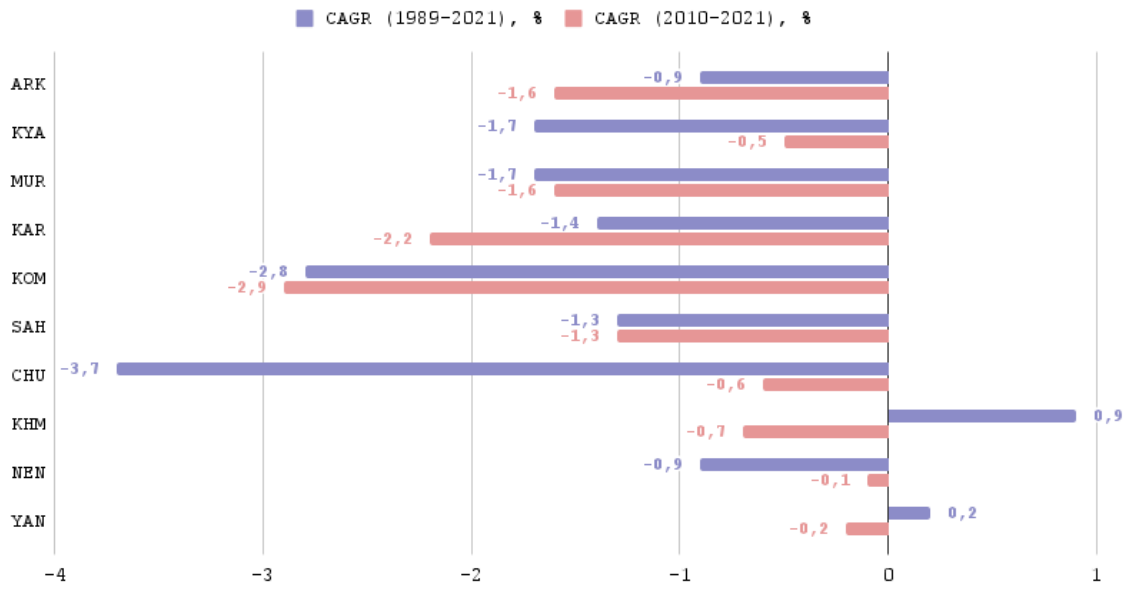
Source: author's work based on governmental database of municipal indicators

Figure 4 illustrates the Compound Annual Growth Rate (CAGR) for Arctic regions, which reflects the average annual dynamics of population during a specific period (see the formula in Appendix 1). In this context, a change of less than –0.15 per cent annually is considered as a shrinkage reference. The CAGR for the entire Arctic between 1989 and 2021 amounted to –1.1 per cent, with approximately half of the Arctic regions exhibiting even lower rates. The CAGR from 2010 to 2021 further deteriorated, reaching –1.28 per cent, indicating a more pronounced decline in the Arctic's population.

The severely shrinking Arctic regions in the recent intercensal period are those located in the northwestern part of Russia (ARK, KAR, KOM). It is noteworthy that at present, 9 out of 10 Arctic

regions are experiencing shrinkage, with the NEN experiencing stagnation with a slight decline. Even in the most actively developing oil and gas regions, such as KHM, NEN, and YAN, the capacity for growth is gradually diminishing. This indicates that in the foreseeable future, the Arctic's shrinkage is likely to continue.

Figure 4. CAGR in Arctic regions / Arctic parts of regions by two periods, %



Source: author's work based on population census data for 1989, 2010, 2021

As it was already noted, the Arctic shrinkage is an urban shrinkage. CAGR was also calculated for the time periods 1989–2021 and 2010–2021, covering all Arctic cities and two strategic settlements that did not have city status (Table 1 in Appendix 2). 4 of the 5 closed administrative-territorial entities in the Murmansk Oblast' were excluded from the analysis due to a lack of available data, leaving a total of 34 shrinking cities under consideration.

It is evident from the data that YAN cities experienced significant growth during the period 1989 to 2021. Out of the total number of Arctic cities, 9 (21 per cent) showed growth. However, one city, Novyy Urengoy, remained stagnating. All other cities were shrinking severely, except for Kostomuksha in KAR. The most pronounced decline was observed in the cities of the KOM and CHU, reflecting the impact of migration outflows.

During the intercensal period from 2010 to 2021, the prevailing tendencies have persisted. However, in recent times, cities of the YAN have experienced population decline, while cities in CHU have demonstrated stability in their demographic dynamics. One particular city, Pevek, has even witnessed overall growth. These observations underscore the fluctuating nature of urban development in the Arctic. Overall, population decline in most Arctic cities remains.

3.2 Analysis of Media Publications and Strategic Planning Documents

The research methodology is rooted in the conceptual framework given in the Literature Review. It is contended that policy as discourse encapsulates certain issues and, through this representation, provides the foundation for policy responses. The analysis of policy as discourse means the examination of policy texts. These texts encompass a wide range of materials, including laws, official documents, political programs, interview transcriptions, media articles, press releases, etc. (Goodwin, 2011).

Anticipated data include media publications featuring statements from policymakers regarding urban shrinkage policies, as well as strategic planning documents from the Russian Federation, Arctic regions, and municipalities experiencing urban shrinkage, providing a multilevel perspective on urban shrinkage policies. The method used is qualitative content analysis (QCA).

3.2.1 Media Publications

Data collection from news media publications is aimed at getting official statements about the Arctic development in the context of population decline between 2014 and 2024. This temporal frame is primarily dictated by the process of establishing a strategic planning framework in Russia (The State Duma of the Russian Federation, 2014). The decade is divided by the year 2020, which marked the approval of state policy in the Arctic region (President of the Russian Federation, 2020). The ultimate objective is to delve into the discussions surrounding urban shrinkage in the Arctic.

The data is uploaded via the Medialogia[2] platform. Building on the empirical assertion regarding the prominent role of governors at the municipal level in Russia (Chirikova & Ledyaev, 2017), the study focuses on the governors of Arctic regions as primary policy transmitters. The data encompassed the timeframes of 2014–2019 and 2020–2024, accounting for the turnover of governors.

The context of news publications is set by word combinations describing population decline: ‘umen’sheniye’ and ‘snizheniye chislennosti naseleniya’ *(population decline),* ‘sokrashcheniye naseleniya’ *(decrease), ‘depopulyatsiya’ (depopulation),* ‘ubyl’ naseleniya’ *(population decline),* ‘migratsionnaya ubyl’’ *(migration loss),* ‘estestvennaya ubyl’’ *(natural loss),* ‘ottok naseleniya’ *(population outflow),* ‘padenie chislennosti naseleniya’ *(fall),* and szhatie *(shrinkage).* During the first timeframe, a total of 1,329 publications were identified, with reprints accounting for 49 per cent of the total. In the subsequent timeframe, 2,579 publications were found, with reprints

[2] ‘Medialogia’ is an automatic media monitoring and analysis system. URL: https://www.mlg.ru/

constituting 48 per cent of the overall count. The analysis encompassed publications from the top 100 media entities, without delving into specific sources, as this could worsen the sample.

All original publications are checked for relevance to the topic, and non-substantive news is eliminated. Non-substantive news includes instances where population decline is merely mentioned without undergoing discussion, or when the messages originate from scholars, experts, or other public speakers rather than officials. Publications on the rural decline or depopulation of non-Arctic territories are also classified as non-substantive.

The resulting data corpus consists of 106 news media publications[3], with 11 originating from federal and 95 from regional actors. The distribution of data across each year within the 2014–2024 timeframe exhibits evident disparities. Overall, there has been a notable increase in interest in the topic since 2020, with a peak in activity observed in 2024.

Table 2. Number of news media publications in the dataset

2014	2025	2016	2017	2018	2019	2020	2021	2022	2023	2024
6	4	7	8	2	5	13	12	4	14	31

Source: author's work

The analysis of the textual data was conducted through qualitative content analysis (QCA). Further details on the methodology can be found in Schrierer (2012) and Elo & Kyngas (2008). QCA employs coding techniques to analyse relatively large volumes of data (Alejandro & Zhao, 2024). The units of analysis are sentences consisting of one thought, numbered from 1 to 3. The analysis was conducted deductively, employing the qualitative coding tools provided by ATLAS.ti[4]. The data was collected by highlighting codes corresponding to the categories: urban shrinkage problem representation, proposed policy responses, and policy actors. These categories propose a framing framework for the QCA. All codes were distributed according to the year of publication.

Problem representation codes and policy responses codes were analysed to identify secondary codes (enlarged codes) and then to form final discourses (themes). Policy actors were also divided into groups based on their territorial level. All information was interpreted in terms of strategic planning and policy formulation.

3.2.2 Strategic Planning Documents

The discourse is also explored in socio-economic development strategies, as well as in state housing and economic programs. According to Federal Law No. 172-FZ, strategic planning documents

[3] The collection of news media publications is available. URL: https://disk.yandex.ru/i/gwKGOuPkOW3gqQ
[4] ATLAS.ti is a software for qualitative analysis. URL: https://atlasti.com/

include: sectoral strategies of the Russian Federation, Spatial Development Strategy of the Russian Federation, state programs of the Russian Federation, socio-economic development strategies and state programs of entities of the Russian Federation, socio-economic development strategies and municipal programs of municipal entities (The State Duma of the Russian Federation, 2014). Strategies set goals and define development indicators, while programs implement specific interventions and financing measures to achieve goals.

In order to study multilevel policy in the Arctic, a total of 113 documents from the federal, regional, and municipal levels have been reviewed.

The federal-level set of documents includes 15 units:

2 Spatial Development Strategy of the Russian Federation (current and previous versions);

4 Socio-Economic Development Strategies of Federal Districts that include Arctic regions;

2 Fundamentals of the State Policy of the Russian Federation in the Arctic (current and previous versions);

2 current and previous Socio-Economic Development Strategies of the Arctic Zone of the Russian Federation;

2 State Programs for Socio-Economic Development of the Arctic Zone of the Russian Federation;

The current Construction and Housing Development Strategy of the Russian Federation;

The current State Program on Housing of the Russian Federation;

The current State Program on Economic Development of the Russian Federation.

The regional-level set of documents includes 49 units:

11 Socio-Economic Development Strategies of Federal Entities of the Russian Federation. In addition to the 10 existing regional strategies, the Republic of Sakha (Yakutia) has a separate strategy for the development of its Arctic territories;

7 previous Socio-Economic Development Strategies of federal entities of the Russian Federation. The Republic of Sakha (Yakutia), the Nenets Autonomous Okrug, and the Chukotka Autonomous Okrug have no outdated strategies available;

10 current State Programs on Housing of entities of the Russian Federation;

5 previous State Programs on Housing of Entities of the Russian Federation. The 5 federal entities — the Yamal-Nenets Autonomous Okrug, the Nenets Autonomous Okrug, the Karelia Republic, the Krasnoyarsk Kray, and the Arkhangel'sk Oblast' — do not maintain outdated versions of the documents. Instead, their state programs are not replaced, but rather updated;

10 current State Programs on the Economic Development of entities of the Russian Federation;

6 previous State Programs on Economic Development of entities of the Russian Federation. 4 subjects — the Yamal-Nenets Autonomous Okrug, the Nenets Autonomous Okrug, Karelia Republic, and the Krasnoyarsk Kray — do not have outdated versions of the documents.

The municipal-level set of documents includes 49 units:

30 Socio-Economic Development Strategies of municipal entities. 3 shrinking cities have not been found to have current strategies available.

19 previous Socio-Economic Development Strategies of municipal entities. 14 shrinking cities have not been found to have current strategies available.

The information on the document count is gathered in Table 3.

Table 3. Strategic planning documents sample

Territorial Level	Non-current	Current	Total
Federal	7	8	15
Regional	18	31	49
Municipal	19	30	49

Source: author's work

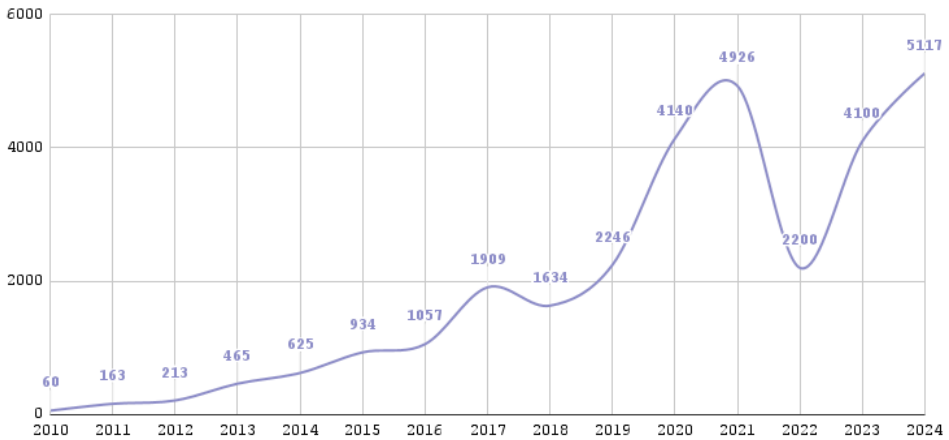
Chapter 4. Results

4.1 The Urban Shrinkage Media Discourse in the Arctic

4.1.1 Description of the Received Data

The mention of the Arctic has been vividly growing. Figure 5 shows the number of mentions of the Arctic Zone of the Russian Federation from 2010 to 2024. There was a sharp increase in mentions from 2019 to 2020 due to the approval of the new Socio-Economic Development Strategy of the Arctic. However, the number of mentions decreased in 2022 but quickly recovered by 2024 due to high media attention on national security.

Figure 5. Frequency of mentions of the Arctic Zone of the Russian Federation



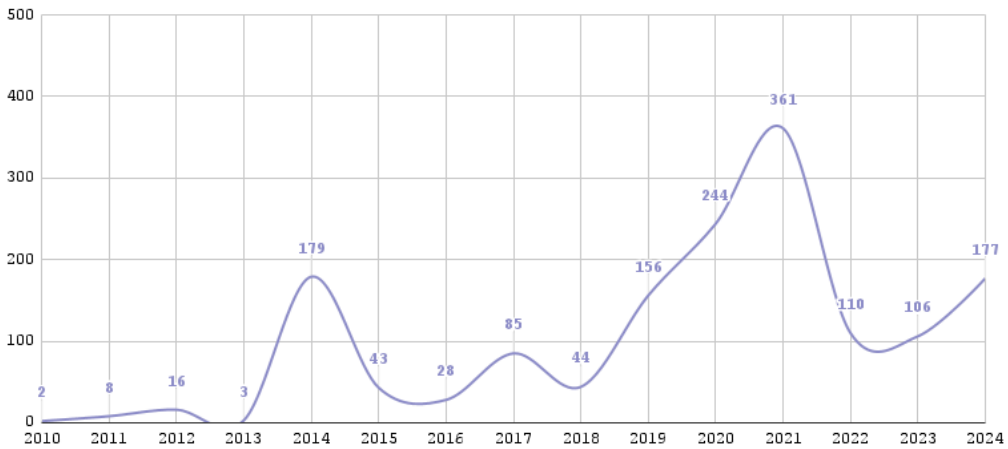
Source: author's work based on Medialogia's data

Figure 6 shows a reference to 'urban' or 'controlled' shrinkage. The spike in mentions in 2014 is associated with an active discussion on single-industry town development in Russia. It was also in 2014 when the classification of these towns based on their socio-economic status was introduced. A slight increase in 2017 can be attributed to the launch of a municipal program for controlled shrinkage in Vorkuta City, Komi Republic. This was when urban shrinkage started to be discussed on a regional level. The growth of mentions between 2019 and 2021 can be linked to an increase in discussion about the challenges of northern shrinking cities in the Komi Republic. Currently, this topic continues to form the agenda for this region.

The role of the Komi Republic is presented in Table 3. In particular, the development of Vorkuta is discussed in terms of urban shrinkage. This discussion seems to imply that urban

shrinkage is a local issue that does not affect other regions. However, it only reinforces the need for a more comprehensive understanding of the phenomenon.

Figure 6. Frequency of mentions of the ‘urban shrinkage’ or ‘controlled shrinkage’



Source: author’s work based on Medialogia’s data

Table 4. Frequency of mentions of words in the dataset on urban shrinkage

Vorkuta	Shrinkage	City	Development	Town	Population	Single-industry town	Komi	Controlled
537	288	264	240	210	179	155	138	128

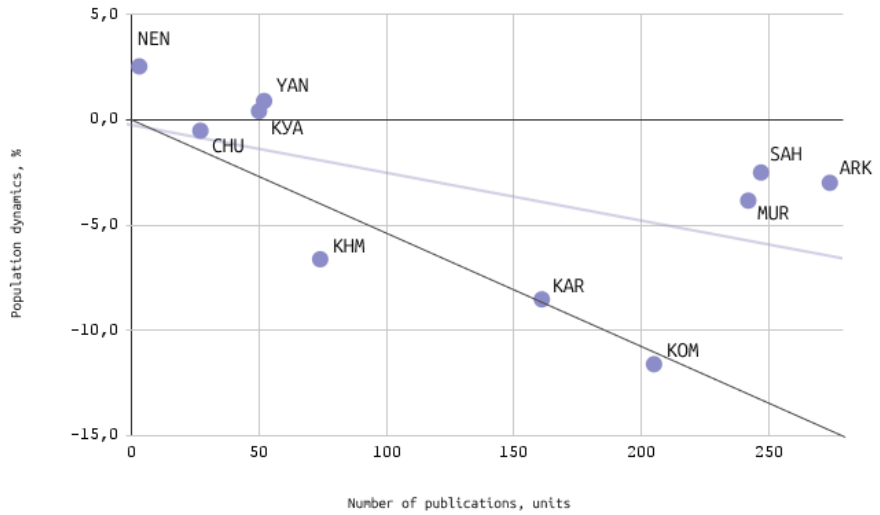
Source: Medialogia’s data

For a study, statistical data on the number of publications, including reprints, related to each of the ten governors for the periods from 2014 to 2019 and 2019 to 2024 was compiled. The context for these publications is the population decline in the respective regions. The findings are significant: in five out of ten regions discussions regarding population decline have become increasingly prevalent over the past decade. These regions include the Murmansk Oblast’, the Sakha Republic (Yakutia), the Arkhangel’sk Oblast’, the Komi Republic, and the Karelia Republic. Tables 1 and 2 in Appendix 3 provide details on the number of reports each year for each Arctic governor. It is worth noting that there has been a notable increase in discussions about population decline within the Arctic region over time.

Figures 7 and 8 present a comparative analysis of the ratio between the number of publications made in 2014–2019 and 2020–2024 and population dynamics in the Arctic regions over the same periods. Ideally, the regions should line up diagonally relative to each other, where the number of publications increases with the magnitude of the population decline. While nine out of ten regions experienced a population loss over time, the publication number varied. In particular,

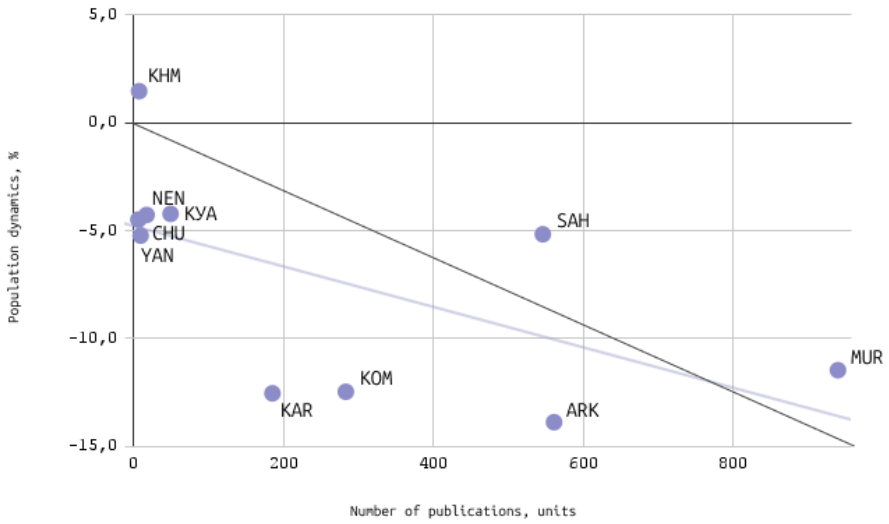
the number of publications increased in ARK, MUR, KOM, and KAR, whereas it decreased in CHU, YAN, NEN, and KYA. It is worth noting that many regions tended to overemphasise discussions on population decline, whereas now, the majority of regions are underreporting this issue. However, it is important to highlight that SAH and MUR are the most active regions in discussing this topic.

Figure 7. Arctic regions by number of publications on population decline and population dynamics, 2014-2019



Source: author's work based on Medialogia's data, Rosstat data (2014, 2020)

Figure 8. Arctic regions by number of publications on population decline and population dynamics, 2020-2024



Source: author's work based on Medialogia's data, Rosstat data (2020, 2024)

4.1.2 QCA Results

The final sample of 106 new media publications was analysed for the frequency of mention of various significant words and their derivatives by using Voyant Tools[5]. Table 5 presents words that appeared more than 100 times. As can be seen, in addition to geographical specifics such as the Arctic and the North, the top ten words include sectoral-specific terms like housing, economy, and two types of government — project and process — as well as three levels of decision-making. Shrinkage-related terms primarily refer to living conditions, including housing, construction, investment, infrastructure, displacement, and mortgage. The most commonly mentioned regions are the Murmansk Oblast' and the Komi Republic, which generate discourse about urban shrinkage in the Arctic most of all.

Table 5. Words' frequency in the dataset for QCA

№	Word	Frequency	№	Word	Frequency
1	Arktika *(Arctic)*	748	18	Realizatsiya *(Realisation)*	186
2	Razvitiye *(Development)*	527	19	Sotsial'nyy *(Social)*	181
3	Region	519	20	Investitsii *(Investment)*	170
4	Sever *(North)*	372	21	Komi	166
5	Gorod *(City)*	323	22	Prezident *(President)*	160
6	Zhil'ye *(Housing)*	320	23	Pravitel'stvo *(Government)*	156
7	Proyekty *(Projects)*	314	24	Monogorod *(Single-industry town)*	146
8	Federatsiya *(Federation)*	291	25	Molodezh' *(Young people)*	143
9	Programmy *(Programs)*	280	26	Infrastruktura *(Infrastructure)*	141
10	Ekonomika *(Economy)*	269	27	Gubernator *(Governor)*	140
11	Naseleniye *(Population)*	268	28	Usloviya *(Conditions)*	139
12	Stroitel'stvo *(Construction)*	266	29	Ottok *(Outflow)*	138
13	Respublika *(Republic)*	253	30	Gosudarstvo *(State)*	131
14	Vorkuta (*Vorkuta city*)	218	31	Munitsipalitet *(Municipality)*	126

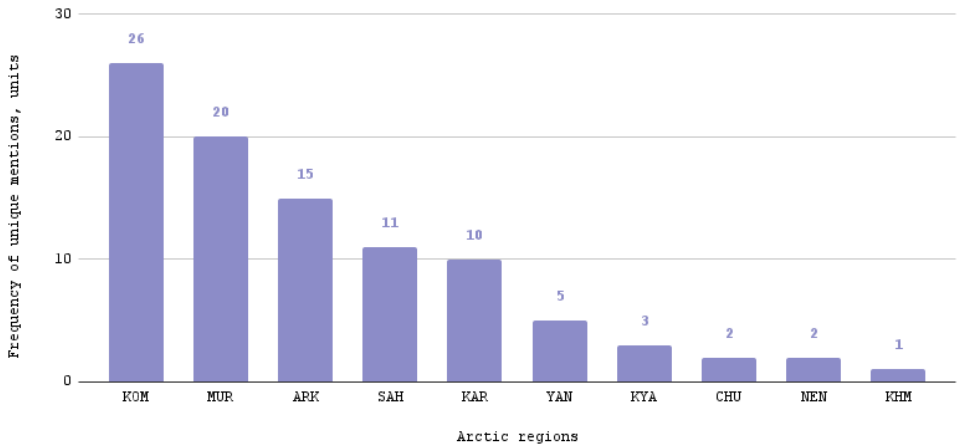
[5] 'Voyant Tools' is a web space for reading and analysing digital texts. URL: https://voyant-tools.org/

15	Murmnskaya *(Murmansk)*	207	32	Byudzhet *(Budget)*	112
16	Podderzhka *(Support)*	206	33	Pereseleniye *(Displacement)*	110
17	Problemy *(Problems)*	202	34	Ipoteka *(Mortgage)*	107

Source: author's work

This statement is confirmed by the frequency of unique mentions of the Arctic regions in the dataset. Figure 9 shows the distribution of these mentions and, in fact, the number of publications originating from each region that were included in the research. In essence, the predominance of publications originating from KOM, MUR, and ARK within the sample aligns with the broader pattern of publications on population decline. This indicates the relative representativeness of this sample for qualitative analysis.

Figure 9. Frequency of the unique mentions of the Arctic regions, units



Source: author's work

During the QCA, textual data were coded into three categories: problem representation, police responses, and policy actors. Table 6 consists of the quantitative characteristics of these categories, specifically indicating the number of initial coding units. A total of 331 initial codes were identified, and 310 actors at various territorial levels were highlighted. The subsequent sections will present a comprehensive analysis of each category.

Table 6. Categories' frequency in the dataset for QCA

Coding units	Problem representation		Solutions		Policy actors	
Number of initial codes	2014-2020	2020-2025	2014-2020	2020-2025	2014-2020	2020-2025
	41	77	57	156	95	215

Source: author's work

Problem representation

In this section of the analysis, the responses of policymakers to some questions were considered:

(1) Why is shrinkage occurring? 105 potential reasons were considered and reduced to six enlarged codes.

(2) How exactly does the shrinkage process go? 70 points were considered and categorised into three enlarged codes.

(3) Which population groups are involved? 22 were found and consolidated into three enlarged codes.

(4) What are the implications of population decline? 57 potential consequences were identified and grouped into three enlarged codes.

Furthermore, all initial codes were assessed in terms of their attitude on population decline: either neutral, where the issue is simply stated without any evaluation, or negative, where population decline is viewed as a catastrophic trend.

Why is shrinkage occurring? The reasons for urban shrinkage in the Arctic regions are divided into Political, Economic, Demographic, and Climate reasons, Infrastructural and Housing problems. Political reasons include the territorial politics of previous regimes in Russia. Economic reasons are linked to economic downturn and deindustrialisation, which have led to a lack of jobs. Demographic factors include population reproduction in the Arctic and the deterioration of human capital due to outmigration and high mortality rates. Climate factors include living conditions in the Far North. Infrastructural problems arise from deficiencies in social infrastructure, urban environments, and transportation, while housing problems stem from a shortage of available housing in the Arctic. Table 7 provides information on the contributing factors. It is of a note that infrastructure and housing problems are associated with urban environments. Thus, the population decrease in the Arctic is caused by the poor living conditions in cities, which discourages people from moving there.

Table 7. Categories of urban shrinkage reasons

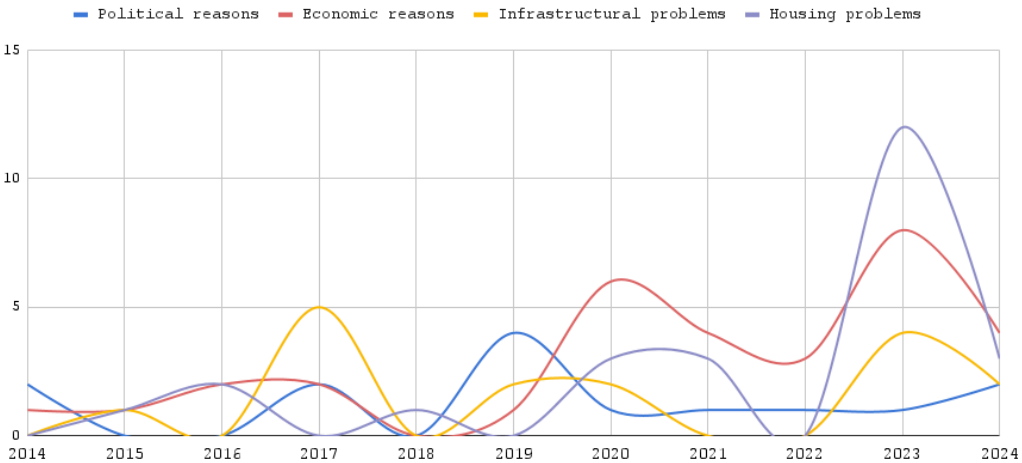
	Political reasons	Economic reasons	Demographic reasons	Climate reasons	Infrastructural problems	Housing problems
Share, %	13	31	8	9	15	24

Source: author's work

Figure 10 illustrates the dynamics of the population decline causes (demographic and climate reasons are excluded, as they are not indicative). As can be seen, from 2019 to 2023, economic issues dominated, mainly due to a lack of employment opportunities, low wages, limited

specialist markets, and low career prospects. Housing problems became more prevalent in 2023 due to preferential mortgage programs for the Arctic region. Although it is hoped that more affordable housing would reduce population outflow, the creation of such housing requires new construction in the Arctic.

Figure 10. Dynamics of shrinkage reasons mentioning, units



Source: author's work

How exactly does the shrinkage process go? A vast majority of codes, accounting for 89% of the total, employ the term 'outflow' or 'migration loss' to describe this phenomenon. The remaining portion either refers to natural loss or lacks any specific explanation. This trend is likely attributed to the historical context of population dynamics in the Russian Far North. Consequently, despite the presence of a growing trend towards natural decline in some Arctic regions, policy responses are more likely to be directed towards the housing sector rather than the healthcare sector.

Which population groups are involved? 68 per cent of all positions are related to young people — this is precisely the group that contributes to the outflow of population and must be retained in the North. Young professionals are sometimes referred to as a separate category, which governors seek to attract to the Arctic region. It is also noteworthy that newcomers are mentioned — they came to work in the Arctic during the Soviet period and migrated back to their home regions. Additionally, there is mention of shift workers in this context.

> The first thing to pay attention to is the conditions necessary to ensure the well-being of young people in the [Karelia] Republic and reduce their outflow from the region. Young people should be encouraged to stay and contribute to the development of the

> human capital and economy of their home territory. [Head of Karelia Republic, 2023].
>
> We do not want to be a region of shift workers — then the region will not thrive, and people will not treat the territory as their own. We can not develop and retain the Arctic only with the efforts of shift workers [Governor of the Murmansk Oblast', 2021].

Despite the Governor of the Arkhangel'sk region reporting that the largest outflow occurs among older people, it is the younger generation that will be the key to maintaining the demographic balance in the Arctic, given the significant population decline. The focus on attracting young people is a strategic move aimed at addressing the threat of population extinction.

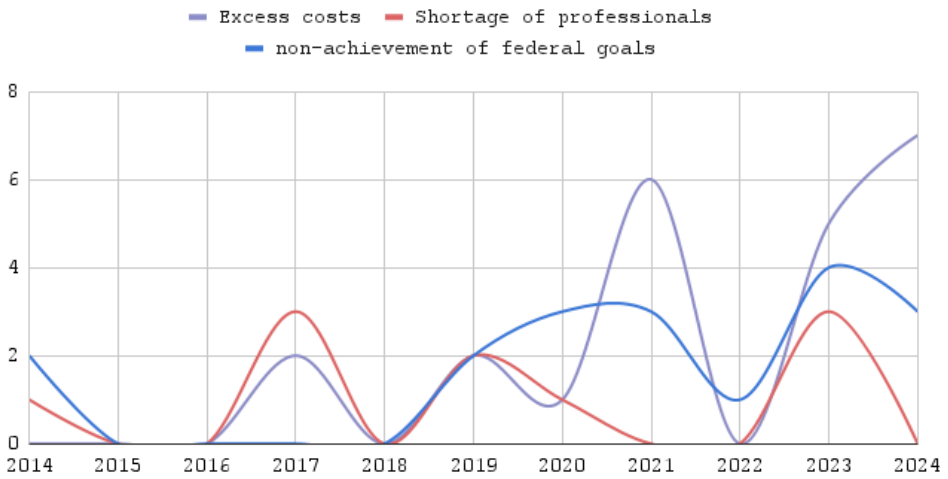
What are the implications of population decline? Three categories were identified, and the dynamics of mentions of positions from these groups is shown in Figure 11.

(1) Vacant housing and infrastructure overcapacity, which result in excessive municipal expenditures. It is a pressing concern in the Komi Republic. The population loss inevitably places a strain on municipal budgets, while excess expenditures are allocated to the regional budget. It is imperative to implement measures aimed at reducing these expenses.

(2) A shortage of labour resources, particularly in projects operating in the Arctic region and industrial sectors. Qualified workers are in high demand for these endeavours. To address this issue, it is crucial to implement strategies aimed at supporting the economy, such as retaining and training local specialists, as well as attracting new people. Additionally, there exists a dearth of professionals in the fields of healthcare and education, necessitating their active engagement through various means.

(3) Failure to meet the indicators and requirements of federal policy. This is an important consequence for officials, not for the territory. The need to address population decline is driven by political considerations. While this may not be explicitly stated, it is essential to achieve the objectives of national projects, foster the development and maintenance of the Arctic under new geopolitical circumstances, uphold national security, and initiate megaprojects.

Figure 11. Dynamics of policy necessity reasons mentioning, units



Source: author's work

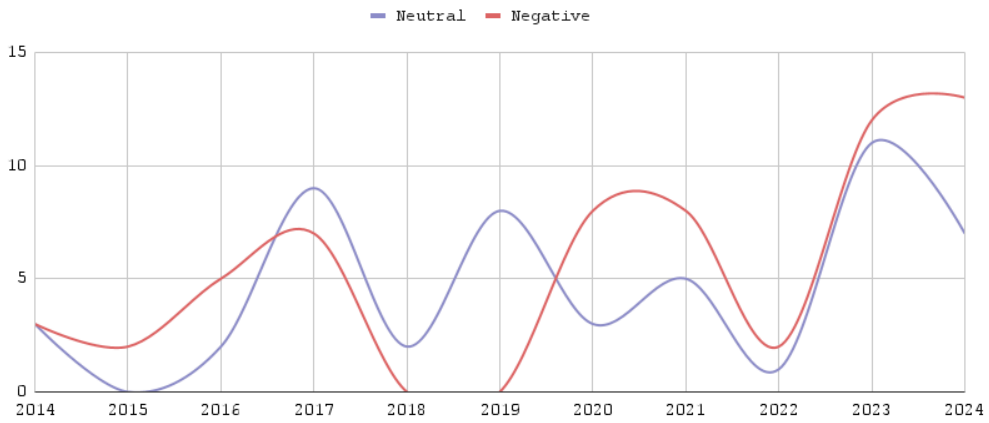
It is not surprising that references to ideological factors have been on the increase since 2019. The need to generate population growth in the Arctic was highlighted in conjunction with the initiation of national projects in Russia in 2018 and the approval of the Socio-Economic Development Strategy of the Arctic Zone in 2020. In parallel with a discourse on reversing population loss, there is an active movement towards adapting to the shrinkage. These contrasting viewpoints are represented by the Murmansk Oblast' and the Komi Republic.

> We are striving to ensure that the number of people leaving the North is as low as possible, while the number arriving is as high as possible. In 2019, we set a goal to reverse the trend of population decline by the year 2030 [Governor of the Murmansk Oblast', 2023].

> The initiative of the Komi Republic is aimed at reducing vacant housing, which has arisen in connection with both the natural and migratory decline in single-industry Arctic towns such as Vorkuta and Inta [Head of the Komi Republic, 2024].

The coexistence of two distinct discourses is further evidenced by the dynamics of urban shrinkage representation, as depicted in Figure 12. The decline observed in 2018 and 2022 can be attributed to the scarcity of publications during those years. However, the multifaceted dynamics from 2019 to 2020 warrant particular attention. Despite a prevailing negative perception of urban shrinkage, with a ratio of 64 negative to 53 neutral views, it cannot be asserted that policymakers regard population decline as a disaster.

Figure 12. Changes in the style of shrinkage representation, units of codes



Source: author's work

Policy responses

A total of 213 initial codes were initially identified, which subsequently expanded to 247 as certain text fragments frequently encompassed multiple distinct responses. These 247 codes have been categorised into five groups.

(1) Social sphere: includes a diverse range of benefits to the general population. This encompasses efforts to attract medical professionals, educators, and social workers, enhance educational opportunities, establish specialised medical standards, and provide social services of a high quality.

(2) Regulatory measures: include the approval of regulatory documents, and strategies, the creation of departments and institutions, competitions for federal funds, and the protectionist policy of the state.

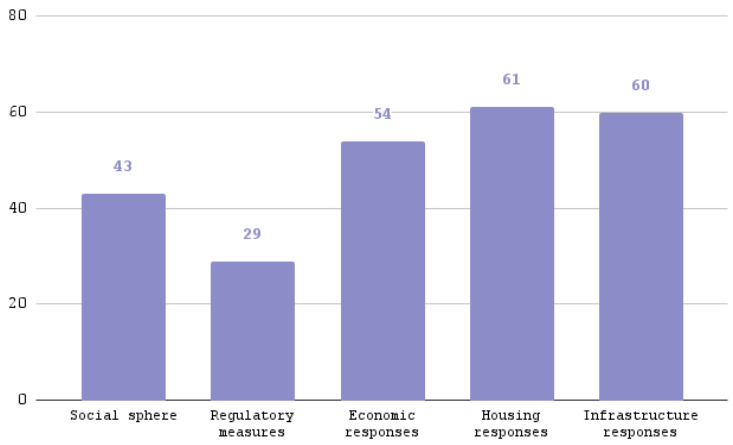
(3) Economy: includes bolstering key sectors of the economy, empowering entrepreneurs, initiating investment projects, fostering job creation, enhancing salaries, and establishing a retraining program tailored to the specific requirements of the Arctic region.

(4) Housing: includes the construction of new houses and attracting developers to the areas for this purpose. It also proposes efforts to reduce the housing fund and increase population density, as well as efforts to resolve housing problems through preferential mechanisms.

(5) Infrastructure: encompasses both the development or renovation of infrastructure and its reduction. Infrastructure is presented in close correlation with the discussion on the comfortisation of Arctic cities. Comfort must be attained through both infrastructure expansion and shrinkage.

As illustrated in Figure 13, both the Housing and Infrastructure categories of policy responses dominate, with the Economic category also comprising a sizable array of codes. This indicates that the primary focus of efforts to combat shrinkage is anticipated to be directed towards the housing sector and enhancing the built environment.

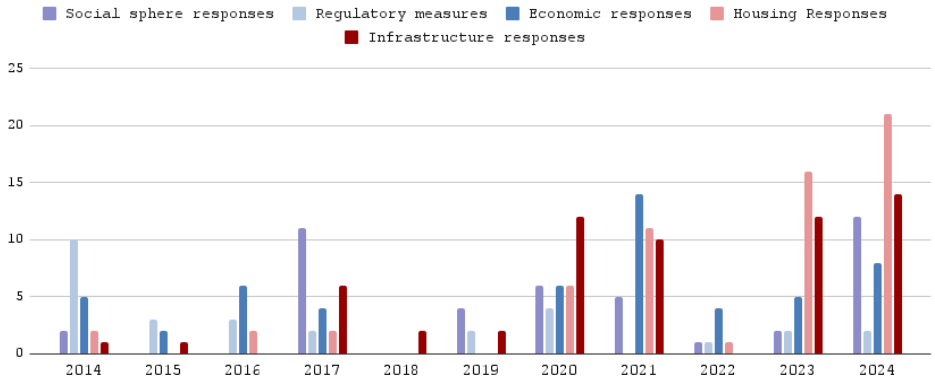
Figure 13. Categories of policy responses, units of codes



Source: author's work

Concerning temporal dynamics, there has been a marked increase in the number of codes in Housing responses and Infrastructure responses categories since the year 2020, as depicted in Figure 14. This sustained growth in the housing sector suggests a heightened focus on the development of urban areas in the Arctic region in response to population loss.

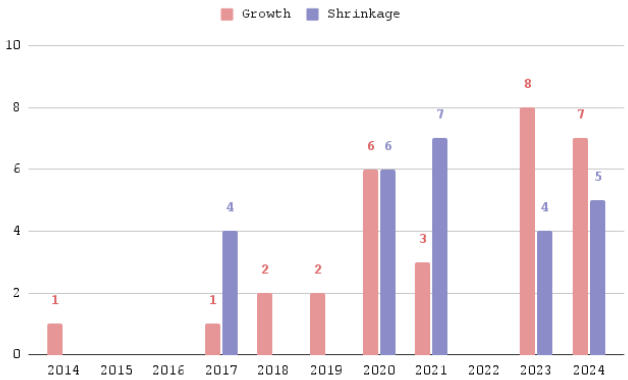
Figure 14. Categories of policy responses by year, units of codes



Source: author's work

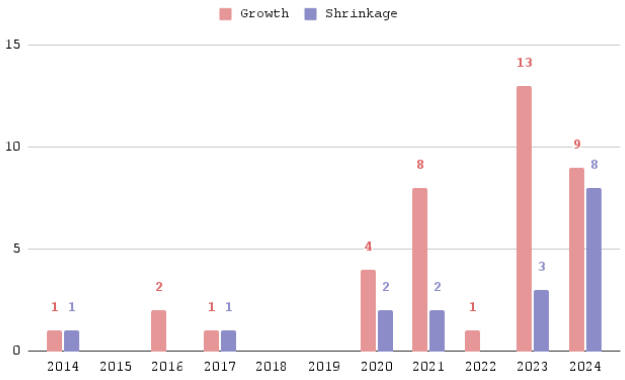
It has been noted that in the most populous categories — Housing and Infrastructure responses — multifaceted policies are evident. Figures 15 and 16 illustrate the distribution of codes aimed at growth and shrinkage in these two domains. In both cases, there is a predominant focus on growth, particularly in terms of infrastructure expansion and housing construction. However, within the realm of housing, the emphasis on fostering growth in Arctic housing initiatives is particularly pronounced. In the infrastructure domain, measures promoting shrinkage account for 46 per cent of the total, and in the housing sector — 34 per cent.

Figure 15. Infrastructure responses, units of codes



Source: author's work

Figure 16. Housing responses, units of codes



Source: author's work

Among the reasons for urban shrinkage, economic considerations related to personnel shortages take precedence. In response, measures in housing and infrastructure areas are prioritised. According to officials' reasoning, economic activity is expected to emerge once favourable living conditions in urban areas are established. Conversely, in the absence of economic attractiveness,

developers are less likely to enter the region, leaving individuals unmotivated to relocate even with the availability of new housing options.

The question of whether the economy or urban environment is more fundamental has not been definitively answered. On the one hand, an economically favourable environment should serve as an impetus for urban development. On the other, it is essential to construct new housing to attract individuals to the Arctic, enabling them to contribute to the region's economic growth. This conundrum is exemplified by statements from officials in Arctic regions.The Komi Republic prioritises economic engagement as a catalyst for Arctic development, with environmental considerations taking a secondary role.

> The creation of conditions for the development of both large and small businesses in the Arctic will provide a new impetus for growth. Incoming investments will allow for the expansion or establishment of new production facilities. This will create new jobs and help reduce the population outflow, directly impacting the social services development and the quality of life improvement [Komi Republic official, 2024].

On the contrary, in the context of federal policy, as well as in other regions actively fighting the issue of population outflow, significant emphasis is placed on comfortisation of urban space and living conditions. This trend is not unique to the Arctic, but it is important that creating a comfortable environment is not merely a means to foster economic development, but also a crucial factor of attracting population to a particular territory.

> We have set the task to create a competitive university, which is the centre of competence [in the Arkhangel'sk Oblast']. This is an ambitious task. Its solution requires the involvement of appropriate teaching staff and specialists. In the current state of Arkhangelsk, this task is particularly challenging. The city requires significant urban environment improvement [Federal official, 2020].

> We are actively seeking to improve the urban environment. We are investing in modern public spaces, renovating facades, and implementing decorative lighting solutions. Not a single city in our region remains untouched by these transformations. To attract young people to our cities, we must leapfrog into the post-industrial era [Governor of the Murmansk Oblast', 2021].

In the year 2017, there was already an active discussion initiated regarding a unified policy for the Arctic region. The focus was on exploring new avenues for Arctic development through the establishment of support zones. This sparked further conceptual discussions among regional leaders about the region's future trajectory. Two distinct discourses emerged: one advocating for growth, and the other advocating for shrinkage. Already in 2017, efforts were underway to enhance the comfort of Arctic cities. However, it is crucial to note that both growth and shrinkage can contribute to improving the quality of life.

> When we discuss the societal role of government, we are primarily concerned with creating the conditions for human life. Indeed, this process [establishing support zones] provides the foundation for the Arctic regions development, the emergence of new industries, the generation of new jobs, and more comfortable living conditions for people living in northern territories, ensuring that they do not feel marginalised in comparison to those living in central and southern areas [Governor of the Arkhangel'sk Oblast', 2017].

> It is imperative to close the economically stagnant settlements that lack prospects for future development, and compact Vorkuta. Additionally, it is crucial to enhance the conditions in two or three selected towns that hold promise for growth. These areas must be equipped with a distinct infrastructure and robust social support systems, fostering a lifestyle characterised by quality and abundance. People need to understand that they live a high-quality life and they have everything needed [Head of the Komi Republic, 2017].

In order to conduct a critical assessment of the measures aimed at addressing population decline, the enlarged codes were revisited and expanded into discourses (themes) (Table 8). It is worth noting that all the enlarged codes focus on increasing the quality of life for citizens. Nonetheless, at the state level, other objectives related to demographic, economic, and housing indicators are pursued. The development of programmes and plans designed to improve people's living standards ultimately boils down to allocating resources in a more targeted manner.

In the Arctic, there is a debate about retaining and attracting populations to permanent settlements in the northern regions. The analysis reveals that the proposed initiatives are geared towards 'buying people' and providing them with enhanced comfort, rather than, for instance, improving the medical and social services available to the existing Arctic population.

Table 8. Description of enlarged codes

Enlarged codes	Description	Codes	Themes (discourses)	Examples (Quotes)
Social sphere responses	Responses aimed at improving demographics	Education improvement, medicine quality, preferences for social staff, support for young families, support for students	Providing financial benefits	'We propose to pay 2 million rubles to specialists who come to work as health professionals in the Arctic' [Governor of the Murmansk Oblast', 2017] 'Health professionals should be afforded greater preferences in terms of salaries and social benefits, for they treat their patients not in an environment of blooming apples, but rather in the ice and frost' [Governor of the Arkhangel'sk Oblast', 2017]
Regulatory measures	Responses aimed at shaping a more precise policy and its financing	Laws or programs approvement, policy reforms	Providing financial benefits	'The Youth Government of the Komi Republic was formed on the initiative of the Head of the Komi Republic [...] and acts as an advisory body to the Head of the Komi Republic' [Newspaper reporter, 2016]
Economic responses	Responses aimed at population retention for the economic indicators growth	Support for entrepreneurs, special modes, arctic personnel, job security, diversification, public-private partnership	Providing financial benefits	'The conditions created for large and small businesses will become a new catalyst for development in the Arctic. [...] This will provide new jobs and reduce the population outflow' [Komi Republic official, 2024]
Responses in Housing and Construction	Responses aimed at rebooting the economy or solving acute problems	Construction of new housing, preferential mortgages, support for developers,	Providing financial benefits Compactisation	'The implementation of the Arctic Mortgage program in the Nenets Autonomous Okrug will contribute to enhancing housing conditions for

		reduction of vacant housing		young families. And not only that! We plan to attract young specialists to our region' [Governor of the Nenets Autonomous Okrug, 2023]
Infrastructure responses	Responses aimed at increasing the comfort of life	Comfortable urban environment, accessible social infrastructure, accessible utilities	Comfortisation	'The factor of [population] attraction and normal living conditions, a comfortable urban environment, and an intellectual environment [Governor of the Krasnoyarsk Kray, 2019]

Source: author's work

Policy actors

While studying urban shrinkage policies, it becomes imperative to pay attention to the policy actors who address the challenge of population decline. In this context, a unique mention of actors of different levels was calculated.

(1) Federal actors: President, representatives from state corporations, and executive authorities such as ministers and heads of departments, legislative authorities such as senators and deputies. The total number of mentions related to these actors amounts to 117;

(1) Regional actors: heads of the Arctic regions, representatives of regional executive and legislative authorities. There are 112 mentions in total;

(2) Municipal actors: heads of the Arctic municipalities, representatives of local administrations, as well as deputies of local councils. The total number of mentions is 37 mentions;

(3) Other actors: industrial and consulting companies, public organisations, educational and scientific institutions. There are 44 mentions in total.

Table 9 presents a policy actors distribution across the timeframes under consideration. The increased prominence of actors from 2020 to 2024 can be attributed to the substantial volume of publications on population decline during this period. Notably, the most significant increase in references occurred at the federal level, which may be directly linked to the intensification of centralised federal policy in the Arctic region, as well as the enhanced role of federal political figures in regional and municipal settings, which emerged following the policy shift towards governance through national projects initiated in 2018.

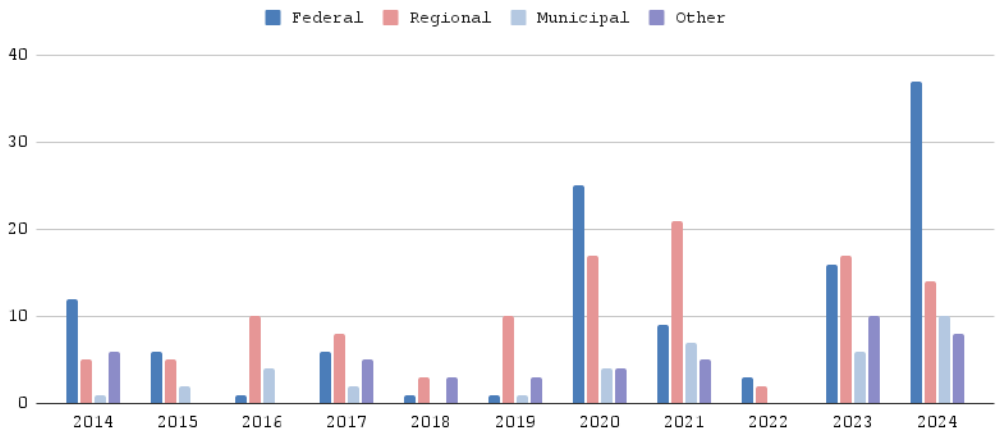
Table 9. Categories of policy actors, units of codes

Federal		Regional		Municipal		Other	
2014-2019	2020-2024	2014-2019	2020-2024	2014-2019	2020-2024	2014-2019	2020-2024
27	90	41	71	10	27	17	27

Source: author's work

Figure 17 illustrates the dynamics of various actors mentioned in the news media publications dataset. Since 2020, there has indeed been a rise in the prominence of federal authorities. The marked increase in references to federal actors in 2024 can be attributed to the initiative of the Komi Republic to introduce a law on preserving abandoned houses, which garnered significant attention from various federal ministries, the State Duma, and the Federation Council. The references to municipal actors have slightly increased recently, owing to the growing visibility of the shrinking Vorkuta city on the federal media stage.

Figure 17. Frequency of the unique mentions of the policy actors, units



Source: author's work

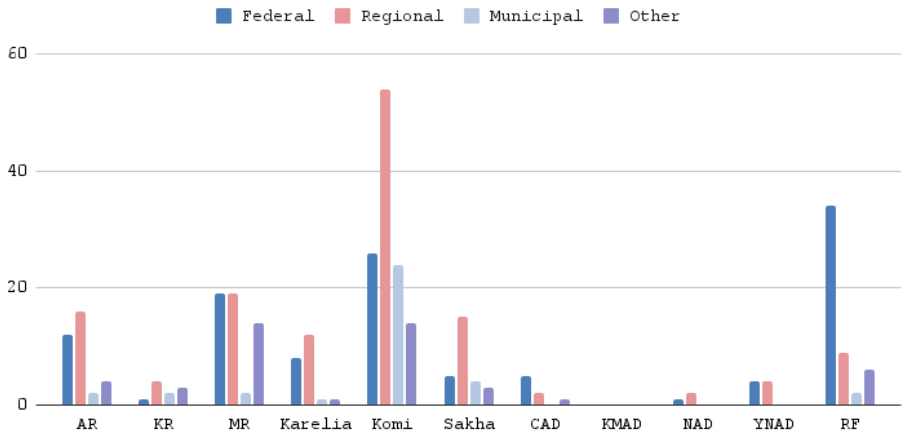
The distribution of actors by Arctic regions, as depicted in Figure 18, is noteworthy. The Komi Republic exhibits the highest concentration of references to regional actors, with discussions on urban shrinkage in the North being led by regional government officials, including representatives from the Ministry of Construction, Housing, and Utilities, the Ministry of Finance, the Ministry of Economics, and the Ministry of Natural Resources; Deputies of the State Council of the Komi Republic and others.

The head of the Komi Republic accounts for only 30 per cent of all mentions, despite being the starting point for sample formation. Additionally, federal actors receive the most attention in the Komi Republic, which can be attributed to the region's engagement with the federal government in

legislative reforms. Furthermore, most of the municipal actors mentioned must be concentrated in the Komi Republic, particularly the authorities of Vorkuta. Overall, the issue of urban shrinkage here is primarily a matter of active regional government efforts.

In the Murmansk Oblast' federal, and regional authorities, as well as private actors, mainly industrial enterprises, all play significant roles. The governor's influence is particularly notable, accounting for 89 per cent of references in a count of regional actors. Consequently, addressing urban shrinkage in this region necessitates the promotion of regional interests at the federal level and fostering public-private partnerships. In particular, companies like Noril'sk Nikel' and Fosagro are encouraged to invest in urban infrastructure improvements. A similar scenario is unfolding in the Arkhangel'sk Oblast' and Karelia Republic, where the involvement of governors at the regional level also stands at a high level: 93 per cent and 75 per cent, respectively. The share of a Head mentioned in the Sakha Republic (Yakutia) is 53 per cent.

Figure 18. Frequency of the unique mentions of the policy actors, units



Source: author's work

Certainly, such inferences may be somewhat prejudiced due to the preponderance of publications originating from the Komi Republic. Nonetheless, the increased prominence of regional authorities in media discussions regarding population decline reflects their greater engagement in addressing urban shrinkage issues.

The relative autonomy of the regional authorities from the Head of the Komi Republic contributes to the coherence of a policy. Urban shrinkage was first mentioned in 2016, when it was suggested to relocate the residents of towns surrounding Vorkuta into the core of the city, providing them with accommodation and employment within the confines of the reduced area. Eight years later, the region has advanced to the stage of revising federal legislation to address the challenges faced by the northern cities of the Komi Republic.

> According to him [the Head of the Komi Republic], these funds should not be allocated for citizens relocating to other regions, but rather for resettlement within the Republic. 'The primary objective [...] is to retain people in the region and create an attractive environment for them to remain. This includes improving working and living conditions locally, such as providing opportunities for towns residents to find employment in Vorkuta through the creation of new jobs' [Head of the Komi Republic, 2016].

> We have already put forward a proposal for a law on the preservation of vacant housing, for this is a matter of federal competence. I believe that with the backing of the senators, we shall arrive at a legislative resolution to this issue. [Head of the Republic of Komi, 2024].

Meanwhile, in the Murmansk Oblast', a dichotomy of perspectives on the concept of shrinkage emerged between the former and current governors, significantly influencing the trajectory of regional policy in a relatively brief timeframe. In the year 2019, there was a perception that fighting population loss was not a viable strategy. However, by 2021, it had evolved into a critical priority for the region's development. This transformation can be attributed to the Murmansk Oblast' role as a frontrunner in shaping the agenda of Arctic development.

> There is no need to fight it [population outflow]. People came here [to the Murmansk Oblast'] for some reason, they came to work, to build. There is no such need now, and there is no point in living here just to live here [Governor of the Murmansk Oblast', 2019].

> The key issue on the Arctic agenda for northern countries today is the fight against population outflow, and the Murmansk Oblast' has created a strategic plan 'In the North — to live! 2.0' [Governor of the Murmansk Oblast', 2021]

With regard to the other regions that are most active in media, it is just stated that population decline is awful and must be combated. However, the specific proposals are not articulated in their statements. Thus, we can observe two parallel discourses on population decline in the Arctic, with the more rational approach of the Komi Republic standing out. Nonetheless, this approach is based on addressing financial issues rather than focusing on population concerns.

A critical inference that must be made in light of the interactions between policy actors is that the issue of population decline is characterised by a marked intersectoral and multilevel nature, irrespective of whether the strategy involves growth or shrinkage. In the context of increased political centralisation, particularly evident in the period from 2018 to 2020, responsibility for the development of shrinking cities has been shifted from the local to the regional and federal levels. This shift is evident in the statement by the mayor of Vorkuta, as quoted below:

> Together with the Government of the Komi Republic, we continue to justify to Moscow that the amount of funding for Vorkuta should be increased several times [...] Optimisation and shrinkage programs are not able to achieve significant savings. We need money and investments. We also need new jobs, and new tax agents, so that the city budget grows. The magnitude of this challenge is at the federal level [Mayor of Vorkuta City, 2024]

Urban development under shrinkage processes is often a costly endeavour, which raises the question at the federal level: Why invest in decline, economic stagnation, or low growth when there are opportunities for profitable investment?

4.2 Urban Shrinkage Discourse in Strategic Planning Documents

4.2.1 Quantitative Perspective of Discourse

For quantitative part of analysis, the shrinking words (SW), which were already used when uploading news publication content, were employed. Due to the absence of explicit discourse on urban shrinkage in state programs, only socio-economic development strategies were considered. A total of 75 socio-economic development strategies were examined, and the information regarding these documents is presented in Table 10.

Table 10. Number of socio-economic development strategies by territorial level

	Previous	Current	*Total*
Federal	6	5	11
Regional	7	11	18
Municipal	19	27	46
Total	32	43	*75*

Source: author's work

All 75 documents were coded by population decline mentions. The following terms were employed for this purpose: ottok *(outflow)*, snizhenie, ubyl' *(decline)*, sokrashenie *(decrease)*, depopulyatsiya *(depopulation)*, spad, padenie *(fall)*, and szhatie *(shrinkage)*. 'Decline' and 'decrease' words neutrally describe population loss, while 'depopulation' and 'fall' show more dramatic dynamics. Shrinkage is a specific term used to describe changes in urban space. Additionally, references to 'migration' or 'natural' were considered to determine the predominant nature of the population decline.

A total of 1,673 references (codes) to shrinkage processes were received for quantitative analysis. Federal documents accounted for 108 codes, regional documents for 390 codes, and municipal documents for 1,175. All codes were categorised by territorial level and document status, with the final table provided in Table 1 of Appendix 4.

Descriptive statistics have been provided for each level. Notwithstanding the varying word count, which is contingent upon the specific documents, it is evident that there are two times fewer mentions of words describing shrinkage per document (see Table 11). It can be attributed to the fact that documents previously adopted regarding the socio-economic development of federal districts primarily focused on addressing population decline, particularly in the North-Western District. Importantly, this district encompasses Arctic regions such as Murmansk Oblast', Karelia Republic, Arkhangel'sk Oblast, and the Komi Republic, which significantly contribute to the discourse on population decline in the media sphere. Interestingly, the Fundamentals of State Policy in the Arctic, adopted more than 15 years ago, fails to mention population loss.

Table 11. Descriptive statistics on federal strategies SW mentioning

Territorial level	Status	Number of SW	Mean	Median	Min	Max	Standard deviation
Federal	Previous	32	12,7	13	0	29	11,7
	Current	76	6,4	4	1	19	7,1

Source: author's work

At the regional level, there has been a simultaneous increase in both the overall number of shrinking words and the average number of words per strategic planning document (see Table 12). The highest peak in the number of words was previously observed in the Murmansk Oblast' (56 words), but now it is in the Yamal-Nenets Autonomous District (52 words).

Some of the regions with the most stable demographic situation, such as the Yamal-Nenets Autonomous Okrug, Nenets Autonomous Okrug, and Khanty-Mansi Autonomous Okrug, have

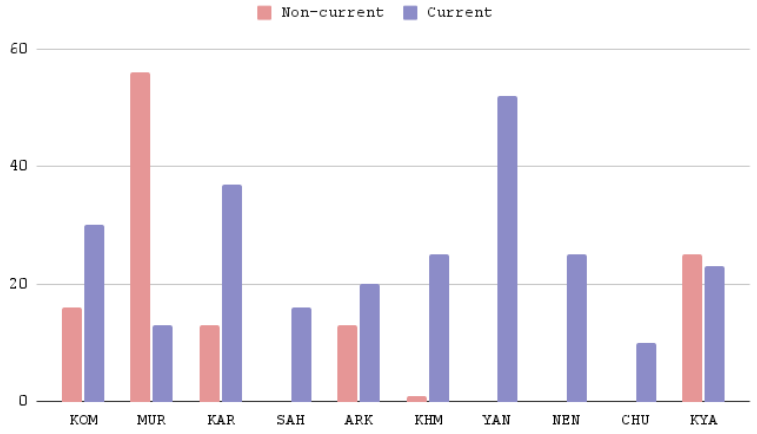
responded to the issue of shrinkage and potential challenges in their socio-economic development strategies. In general, as illustrated in Figure 19, there has been an increase in mentions of shrinkage processes in all regions, except for the Murmansk Oblast' and Krasnoyarsk Kray, where there has been a decrease. The variability in values (standard deviation) remains significant, indicating a high degree of heterogeneity in documents related to population decline.

Table 12. Descriptive statistics on regional strategies SW mentioning

Territorial level	Status	Number of SW	Mean	Median	Min	Max	Standard deviation
Regional	Previous	124	17,7	13	0	56	19,0
	Current	266	24,2	23	10	52	12,1

Source: author's work

Figure 19. SW mentions in regional current and previous strategies, units



Source: author's work

The heterogeneity of documents at the municipal level is even more pronounced, and it has been increasing over time, in contrast to documents at the regional and federal levels. An absolute record for the number of mentions of shrinkage terms can be observed in the socio-economic development strategy of Vorkuta, in the Komi Republic, containing a remarkable 106 references to shrinkage (see Table 13). It is crucial that both in municipalities and in regions, there has been a rise in the frequency of mention of shrinkage processes.

Table 13. Descriptive statistics on municipal strategies SW mentioning

Territorial level	Status	Number of SW	Mean	Median	Min	Max	Standard deviation
Municipal	Previous	409	21,5	18	3	53	14,0
	Current	766	27,4	23	1	106	23,0

Source: author's work

Concerning the mentions of population decline in previous documents, migration decline accounts for 22 per cent of occurrences of shrinkage terms, while natural decline constitutes 8 per cent, as it is presented in Table 14. In the current versions, both indicators have increased by an average of one document. Migration processes continue to describe the population decline in the Arctic region, although the indicator for natural decline has increased to a greater extent.

Table 14. Characteristics of population decline representation

	Migration loss, units	Migration loss, %	Natural loss, units	Natural loss, %
Previous	134	22	51	8,4
Current	258	24,2	121	11,4

Source: author's work

4.2.1 Policy Change Through the Arctic Shrinkage Discourse

For qualitative analysis purposes, goal-setting texts from strategies and state programs were examined. The frequency of specific words used in setting goals was analysed, the discourses on shrinkage were described, and the policy measures contained in state programs designed to address population decline were explored. Table 15 presents the most prevalent words in strategies across different generations, excluding those that appeared fewer than ten times. Notably, there has been a significant increase in the use of terms such as 'comfort' and 'environment'.

Table 15. Frequency of words in goal-setting of Socio-Economic Development Strategies

№	Word	Non-current	Current	Change, %	№	Word	Non-current	Current	Change, %
1	Ekonomicheskiy *(Economic)*	25	30	16	7	**Arktika** *(Arctic)*	**7**	**16**	**56,3**
2	Kachestvo *(Quality)*	20	28	28,5	8	Povysheniye *(Increase)*	20	24	16,7
3	Naseleniye *(Population)*	23	30	23	9	**Komfort** *(Comfort)*	**2**	**13**	**84,6**
4	Razvitiye *(Development)*	29	33	12	10	**Sreda** *(Environment)*	**3**	**16**	**81,2**
5	Zhizn' *(Life)*	21	28	25	11	Ustoychivyy *(Sustainable)*	9	13	30,7
6	Obespecheniye *(Security)*	16	20	20	12	Zhil'ye *(Housing)*	9	11	18,2

Source: author's work

To comprehend the context of population decline discourse in the Arctic, it is crucial to discuss how this issue is represented in high-level policy documents. The new Spatial Development Strategy introduces a specific indicator for the Russian Arctic that aims to maintain the ratio of permanent residents at the level it was in 2023. Additionally, the target is set at 100 per cent by 2030 (Government of the Russian Federation, 2024). This suggests that the population decline in the Arctic must be stopped. There is no such indicator in the previous version of the Strategy.

The issue of population loss in the Arctic region has been previously proposed to be addressed by creating favourable conditions and incentives aimed at reducing migration outflow and attracting professionals to territories experiencing labour shortages (Government of the Russian Federation, 2019). Currently, the emphasis is shifting towards infrastructure development, which is expected to stop the population migration from cities, which play a role as the major service centres towards peripheral areas within the Arctic.

The issue of population decline is also addressed in the current Foundations of State Policy in the Arctic Zone of the Russian Federation (President of the Russian Federation, 2020a). The document underscores that the depopulation of the Arctic constitutes a critical threat to national security. In contrast, the previous document failed to articulate any policy regarding the issue of population decline.

The discourse presented in Socio-Economic Development Strategies of the Arctic has also transformed. Previously, population decline was identified as a crucial factor affecting the

socio-economic evolution of this macroregion. It was particularly emphasised that the outflow of highly skilled labour poses a significant obstacle to the advancement of the Arctic (President of the Russian Federation, 2013). At present, demographic decline represents a significant threat to the sustainability of Arctic development (President of the Russian Federation, 2020b). In the realm of social objectives, the emphasis on goal-setting has shifted specifically in the domains of infrastructure and housing. Whereas previously, efforts were focused on modernising social infrastructure and housing to prevent population exodus, the current emphasis lies in creating a modern and comfortable urban environment, coupled with state support for housing construction in the Arctic.

The Socio-Economic Development State Program of the Arctic Zone has evolved into a more ambitious initiative, shifting its focus from merely enhancing socio-economic conditions to fostering accelerated economic growth. This shift necessitates the creation of new jobs (Government of the Russian Federation 2014b; 2021). The current State Programs on Housing and on Economic Development of the Russian Federation consist of the aforementioned objectives in their goal-setting: the augmentation of the annual quantity of commissioned housing, the enhancement of the urban environment, and the establishment of new jobs to bolster the capacity of regional economies. Additionally, there is a focus on fostering employment within small and medium-sized enterprises (Government of the Russian Federation, 2014a; 2017).

Indeed, all these documents delineate the specific focus in which it is necessary to consider the issue of demographic decline in federal Arctic entities and municipal entities. This paper presents an in-depth analysis of the strategic initiatives and governmental programs implemented by the Arctic regions and their municipalities. The following is an analysis of the strategies and state programs of the regions and municipalities of the Arctic Zone.

Arctic Regions. Previously, population decline presented a significant challenge, primarily in the form of an impending labour shortage in the economy. This issue is highlighted in the strategic documents of the Komi Republic, Murmansk Oblast', Karelia Republic, and Yamal-Nenets Autonomous District. Strengthening demographic and migration policies has emerged as a crucial measure for these regions (Government of the Karelia Republic 2007; Government of the Komi Republic, 2006; Government of the Murmansk Oblast', 2010). On the contrary, the strategy of the Khanty-Mansi Autonomous Okrug emphasises the complete absence of such an issue, as there has been an influx of workers into the region. The Krasnoyarsk Kray separates the northern part from the rest of its territory, and the decline in the economically redundant population in the Far North is a consequence of deliberate policy implemented by the region (Government of the Krasnoyarsk Kray, 2013a).

A more nuanced perspective on depopulation is offered in the contemporary regional socio-economic strategies. The consequences of population decrease have become more accurate: problems with (1) employment in the single-industry towns (Komi Republic, Khanty-Mansi Autonomous Okrug); (2) vacant housing (Komi Republic, Murmansk Oblast'); (3) excess infrastructure (Yamal-Nenets Autonomous Okrug); labour loss (Arkhangel'sk Oblast', Karelia Republic); (4) tension in the provision of social services (for instance, Sakha Republic (Yakutia)).

Consequently, among the measures to be taken, economic diversification stands out, involving both federal and extra-budgetary investments. This includes efforts to reduce the cost of maintaining vacant housing, as well as targeted measures aimed at attracting and retaining human resources (for example, Government of the Murmansk Oblast', 2013a). Particularly, there is a focus on fostering tourism and cultivating a favourable perception of the region (for example, Governor of the Archangel'sk Oblast', 2019).

As can be inferred from the preceding paragraph, the discourse on population decline is largely shaped by the regions that are most prominently represented in media discourse: Komi Republic, Arkhangel'sk Oblast', Murmansk Oblast', Karelia Republic, Sakha Republic (Yakutia). The Yamal-Nenets and Khanty-Mansi autonomous okrugs are also mentioned here. These regions, which have experienced steady population growth for a long period, are now facing increasing challenges in their mono-resource cities. It is noteworthy that the Nenets Autonomous Okrug, Krasnoyarsk Kray, and the Chukotka Autonomous Okrug are not included in this discourse.

The strategies of these regions reveal that for small mining regions such as the Nenets Autonomous Okrug and Chukotka Autonomous Okrug, the constant population fluctuation related to the industrial development of the Arctic is a normal phenomenon. Krasnoyarsk Kray is primarily concerned with the Krasnoyarsk agglomeration in the southern part of the region, where the population is concentrated. The depopulation in the northern areas of this region is seen as a natural process.

Arctic municipal entities. Urban development has previously been viewed primarily through an economic lens in municipal goal-setting: 'The establishment of a competitive economy' (Purovskiy Municipal District Duma, 2011), 'Increasing investment attractiveness' (Segezha Municipal District Council, 2015), 'Ensuring economic growth' (Administration of the Apatity, 2016). Economic measures also prevailed: 'Adaptation [of the economy] to the recession' (Inta Urban District Council, 2014), 'Jobs creation' (Kola Municipal District Council of Deputies, 2015), 'Development of a municipal program to reduce the migrant workers' outflow' (Beloyarskiy Municipal District Duma, 2014).

In the excerpts that represent the issues of local-level shrinkage, the words 'trudovoy' *(labour)* (7 units), 'obrazovanie' *(education)* (7 units), and 'zdravoohranenie' *(healthcare)* (5 units) stand out. This overall pattern highlights both the sectors that are most in need of labour and the underlying issue itself: labour is a crucial component in the advancement of the Arctic cities.

The process of goal-setting at the local level has undergone a significant transformation and expansion, with a shift towards prioritising the urban environment with economic attractiveness. The focus has shifted to 'Creating comfortable living conditions for the population' (Kandalaksha Municipal District Council of Deputies, 2018), 'Creating a comfortable space for the human capital development' (Beloyarskiy Municipal District Duma, 2023), 'The preservation of human potential through the creation of a comfortable living environment' (Turukhanskiy Municipal District Strategy Project, 2020). Thus, comfort also becomes an important factor in population retention.

The measures outlined in the documents provide further confirmation of this trend. Measures aimed at creating a comfortable environment are prevalent, with 15 instances of mention. Support for small and medium-sized enterprises also enjoys widespread recognition, with 16 instances of mention. Healthcare remains a priority, with 9 instances of mention, while education, as a crucial aspect in preserving human capital, has seen a decline, with only 3 instances of mention. This suggests that in shrinking cities, the discourse is shifting towards a focus on fostering a robust economy and creating a comfortable environment, thereby ensuring a high standard of living for the population and preventing its outflow.

The issue of population decline has become more pronounced in recent times, as evidenced by the increased frequency of shrinkage words mentioned. Whereas previously, the population loss was primarily associated with social (demographic) and economic aspects, it has now extended to encompass a broader range of sectors, as exemplified by the housing sector.

Following the analysis of previous generation socio-economic development strategies, the following discourse emerges: *Population decline as a threat to the economic processes termination*. The retention or attraction of the workforce is essential. The population outflow results in a decline in the quality of human capital, necessitating economic interventions that enable both the local workforce to remain in the Arctic and facilitate the integration of newcomers into the region. The analysis of current generation socio-economic development strategies leads to such a discourse: *To stop the population decline at all costs*. This implies a multifaceted approach to address this issue from all possible sectors simultaneously.

Discourse in Housing: Comfortisation Turn for Urban Growth

The indicator of annual housing commissioning in forecasts is not presented in all strategies and state programs of the constituent entities of the Arctic Zone of the Russian Federation (Appendix 5). This fact makes it impossible to determine whether the indicators of development at the subnational level have improved or deteriorated. It should be noted that regional state programs on housing do not take into account population dynamics, irrespective of the approval time of the documents. State programs on housing in the federal entities, as a rule, focus on pragmatic objectives related to relocating residents from unsuitable emergency housing and providing accommodation for various vulnerable social groups. Nevertheless, it is worth paying attention to the goals set and the specific measures proposed for implementation. Housing plays a crucial role in the Arctic region, where poor living conditions contribute to a lack of attractiveness for the population.

Following an analysis of the planning documents of the preceding period, a dominant discourse (1) '*Housing and utility security under extreme conditions*' has been identified. The safety of individuals residing in the Arctic is of paramount importance. This is accomplished through the enhancement of the dependability of engineering infrastructure systems (Government of the Murmansk Oblast', 2013b), increasing the security level of citizens through improving their living conditions (house repairs) (Government of the Chukotka Autonomous Okrug, 2017).

As a result of the analysis of strategic planning documents of the current generation, two enlarged codes were identified:

(1) *The enhancement of living standards through the expansion of construction*. Previously, there were only sporadic indications of the need for housing construction, but now this has become the most important objective of housing policy on the sub-national level.

(2) *Comfort and urban environment improvement as the primary impetus for development.* 'Comfortable housing' and 'comfortable environment' have become firmly established in the policy formulation, particularly in the context of the establishment of the 'Housing and Urban Environment' national project (2018-2024).

This analysis underscores the trajectory of policy at the federal level. Notwithstanding the fact that state programs are devoid of any ideological underpinnings, the discernible discourses indicate a paradigm shift from a focus on security to a prioritisation of convenience. The realm of housing, which once served as a bastion of ensuring basic human necessities, has evolved into a domain of expansion and growth.

> At present, the housing sector in the Murmansk Oblast' is undergoing a period of expansion and advancement. The rate of housing construction is on the rise, and collaborative efforts are being made with developers to foster the construction of apartment buildings [Government of the Murmansk Oblast', 2020a]

In five Arctic regions, specific supporting measures for shrinkage response have been identified. It is remarkable that these measures are designed to address utilitarian issues or provide financial benefits to various social groups with the goal of attracting them to the Arctic.

The Murmansk Oblast' had previously implemented subsidies aimed at compensating for the revenue shortfall experienced by resource-providing organisations (Government of the Murmansk Oblast', 2013b);

The Sakha Republic (Yakutia) had provided financial assistance for the purchase of housing for teachers and medical professionals who have devoted 5 years of their service in the Arctic and remote municipalities organisations of the Republic (The Head of the Sakha Republic (Yakutia), 2017));

The Komi Republic provides financial assistance to municipal entities for the resettlement of underpopulated areas, including the renovation of vacant municipal housing units (Government of the Komi Republic, 2019b). The Komi Republic has previously collaborated with the Ministry of Construction of the Russian Federation to secure federal funding aimed at optimisation of the efficiency of Vorkuta's heating system (Government of the Komi Republic, 2012a);

The Arkhangel'sk Oblast' extended financial support to young families aimed at facilitating the acquisition or construction of residential property within the region (Government of the Arckhangel'sk Oblast', 2013a);

The Yamalo-Nenets Autonomous District grants subsidies for the engineering infrastructure projects development aimed at providing land plots for the construction of individual houses intended for families with three or more children (Government of the Yamalo-Nenets Autonomous Okrug, 2013a).

The measures implemented in the Krasnoyarsk Kray are designed to facilitate the relocation of unviable settlements. Specifically, a subsidy is allocated to the budget of the Taimyr Dolgano-Nenets District to provide social assistance to retirees relocating outside the district for the acquisition or construction of residential property. Additionally, support is provided to the Evenki District for providing social benefits to individuals relocating outside its borders to purchase or construct housing (Government of the Krasnoyarsk Kray, 2013b).

The support measures that have been proposed only serve to reinforce the narratives that were identified during the analysis of media publication: in order to promote the shrinking territories development, individuals must either be enticed with financial incentives, retained through such incentives, or relocated in order to conserve budgetary resources.

Discourse in Economy: New Jobs as an Intangible Remedy Against Shrinkage

As previously stated, the economic sphere continues to be the most significant factor influencing demographic trends in regions and cities. Consequently, there have been no substantial alterations in the discourse on the regional level. Some of the concepts remain fundamental principles: 'Favourable business environment' (Government of the Murmansk Oblast', 2013c), 'competitive industry' (Government of the Komi Republic, 2019c), 'sustained economic growth' (Government of the Arckhangel'sk Oblast', 2019). The economic growth stimulation in territories is a critical area of focus for state authorities in the entities of the Arctic Zone of the Russian Federation. In the context of shrinking regions, this endeavour is directed towards the preservation of the population.

The analysis of media publications has already revealed that job creation is a critical indicator for the government, as it aims to address the issue of population decline in Arctic regions. The emergence of new jobs is expected to result from the diversification of economic activities and the establishment of novel industries, serving both to retain and entice the population.

Nonetheless, in the analysis of state programs of federal Arctic Entities, it becomes evident that the indicator of 'newly created jobs' is not forecasted. Consequently, it is challenging to envision a comparative analysis of quantitative indicators that characterise regional plans for urban development. It is worth noting that in certain regions, previous planning documents do incorporate the estimation of job creation for the forecast period; however, these projections are linked to small and medium-sized enterprises that receive governmental support or to residents of the Arctic Zone (for instance, Government of the Komi Republic, 2012b). Specific indicators are presented only in the Krasnoyarsk Kray: 'The number of high-performance jobs created is 40 units annually' (Government of the Krasnoyarsk Kray, 2013c) and the Sakha Republic (Yakutia): 'Creation of 136,000 new jobs by 2020' (The Head of the Sakha Republic (Yakutia), 2011)).

As a general rule, regions use the job creation indicator as a benchmark for awarding subsidies to local governments to support small- and medium-sized businesses (for example, in the Murmansk Oblast' and the Karelia Republic). While such assistance is crucial for local economies, it does not significantly contribute to the overall economic development of the regions. Consequently, regions primarily exert their influence indirectly.

Particularly noteworthy are the specific measures of economic support implemented in the context of shrinkage in the Komi Republic, Murmansk Oblast', and Chukotka Autonomous Okrug. Of these, the Komi Republic stands out for its systematic approach to addressing shrinkage issues in problem territories.

The Komi Republic grants financial assistance to municipal entities in order to mitigate the effects of the single-industry enterprise closure within resource-dependent municipalities (Government of the Komi Republic, 2019c). Previously, there was also an initiative aimed at assisting in the development or rehabilitation of infrastructure assets essential for the realisation of new investment ventures in single-industry towns (Government of the Komi Republic, 2012b).

In the Murmansk Oblast', there was a program aimed at supporting small and medium-sized enterprises to foster their growth in single-industry towns (Government of the Murmansk Oblast', 2013b).

In the Chukotka Autonomous District, financial incentives were previously provided to attract workers from other regions of the Russian Federation (Government of the Chukotka Autonomous Okrug, 2013).

Chapter 5. Discussion

This chapter is devoted to defining the role of urban shrinkage discourse in multilevel governance in the Arctic Zone of the Russian Federation. The role is determined by the interplay between how the policy is perceived (formulated in news media publications) and how it is subsequently translated into action through strategic planning (formulated in strategic planning documents) at three levels of governance.

The discourses in the media were analysed from a multilevel perspective, revealing how policy regarding the urban population decline in the Arctic region is formulated. One of the key findings is that depopulation is closely linked to external economic factors, however, urban shrinkage problems and responses are concentrated in the housing sector. The distinctive feature of the Arctic context is that the development of its cities is contingent upon market demand for extracted resources and industrial output, resulting in job instability and fluctuating populations.

As can be inferred from a QCA of media publications, to prevent the population from leaving and to attract the needed professionals, it is essential to provide them with unique opportunities that cannot be found elsewhere. For instance, offering them the chance to acquire new housing at an exceptionally low interest rate compared to the national average *(Providing financial benefits)*. Moreover, to maintain a stable population, living standards in the Arctic cities must be at least equivalent to those in the central parts of Russia *(Comfortisation)*. The impulse to develop new housing and infrastructure as a response to the economic downturn may be attributed to the fact that such initiatives are often perceived as more feasible than efforts to make the Arctic region market competitive. This narrative bears a striking resemblance to the Soviet approach, where vast amounts of housing were constructed for migrants who subsequently departed, leaving behind a host of challenges for the Arctic cities.

The discourse on urban shrinkage in the media is characterised by inconsistency and ambiguity. Alongside efforts to create conditions conducive to population growth, there exists a prominent discourse regarding compactisation — the process of condensing a shrinking population to address an acute issue *(Compactisation)*. The prominence of this discourse within the media can be attributed to the promotion of this issue at the highest levels, resulting from impediments in federal legislation.

It is worth emphasising that all these discourses are predominantly shaped by three distinct regions: the Murmansk Oblast', the Arkhangel'sk Oblast' and the Komi Republic. The first two regions are the main competitors in the struggle for a flagship in the Arctic, they include the largest cities in the macroregion — Murmansk and Arkhangel'sk. These regions are associated with

Russia's aspirations to transform the Arctic into a new hub of growth, prosperity, and comfort. The frequent mention of population decline in Murmansk Oblast' is indicative of the regional government's focus on fighting this problem. In contrast, for the Komi Republic, Arctic territories represent the main pain rather than a source of development. The problems accumulated in the most depopulating Arctic cities such as Vorkuta (–3 per cent per annum) and Inta (–4.2 per cent per annum) force the regional government to take an active part in their solution.

The remaining regions are largely eclipsed by the discussion of this issue for a variety of reasons. Demographic decline may be not a critical challenge that necessitates intervention. This is why there is a dearth of news media publications on population loss in regions such as the Nenets Autonomous Okrug, Yamal-Nenets Autonomous Okrug, Khanty-Mansi Autonomous Okrug, and Chukotka Autonomous Okrug. These regions have distinct and more recent histories of urban development compared to the cities in the Murmansk Oblast', the Arkhangel'sk Oblast', the Komi Republic, which have been the subject of extensive media coverage, and the Krasnoyarsk Kray. However, Krasnoyarsk Kray also intervenes in its cities' lives without media attention, and the population decline in Arctic territories is a manageable process in this region.

An examination of the strategic planning documents also reveals that federal strategy establishes a blueprint for action in the Arctic. The primary objective at the federal level is to halt population decline at all costs, as the population is critical to the survival of this macroregion and the security of the nation of Russia as a whole. It is at the federal level that initiatives are proposed to stimulate housing construction in the Arctic and jobs creation for the influx of new residents.

The notable shifts in discourse at the regional level, associated with the expansion of areas of influence on demographic dynamics, can be directly linked to the imperative to stop population decline through any means available. However, regional policies can vary significantly and exhibit remarkable adaptability. This was evident in the cases of the Komi Republic and the Krasnoyarsk Kray, which prioritised the urban shrinkage of Northern areas in their policy frameworks. This suggests that not all regions subscribe to a uniform Arctic agenda set by the federal government, but rather prioritise addressing their specific challenges. Typically, there is no autonomous goal-setting at the local level. Rather, it is contingent upon the objectives set by the regional government and the minor measures it can implement independently. Consequently, the dynamic relationship between discursive policy formulation in news media publications and strategic planning documents is illustrated in Table 16.

Table 16. Comparison of urban shrinkage discourse in media and planning documents

	Media discourse			Strategic planning discourse		
	Problem representation (discourse)	Policy responses (discourse)	The actors' role	Problem representation (discourse)	Policy responses (discourse)	The actors' role
Federal level	Shrinkage is a consequence of **economic factors**, but its impact is particularly acute in the **housing sector**, and the solutions lie primarily within the **housing sector**.	Providing financial benefits Comfortisation Compactisation	Report on priority support measures for the geostrategic region; Set the goal for Governors to reduce the population decline.	The population decline is no longer merely a social issue to the Arctic, but a menace to national security; Stop the population decline at all costs	*Discourse in Housing:* Comfortisation Turn for Urban Growth; *Discourse in Economy:* New Jobs as an Intangible Remedy Against Shrinkage	Create a framework for policy responses; Create federal support measures aimed at accelerating development in the Arctic
Regional level			Generate the most visible discourse on the issue of urban shrinkage; The main policymakers and policy guides to shrinking cities.			(Un) Solves the problems of cities in the context of shrinkage
Municipal level			Policy recipients; Engage in deliberations at forums of various levels on pressing matters			Policy recipients, practically do not adopt
Other actors			Contribute to the master plans development; Engage in educational processes; Establish industrial and civil infrastructure			Act as important actors for the economic well-being of shrinking territories

Source: author's work

The research, based on qualitative data, reveals that the current discourse on urban shrinkage in the Arctic is not representative of the entire region and is predominantly shaped by a limited number of key actors who exert a significant influence on Arctic development. Specifically, this discourse is primarily driven by federal actors, with support from the Murmansk Oblast' and Arkhangel'sk Oblast'. Importantly, there has been a noticeable increase in discourse on population loss over the past decade, leading to the assumption that the entire Arctic region should be guided by a paradigm focused on fighting population decline. However, it is important to acknowledge that Arctic policies are formulated by 10 federal entities that adhere to the prevailing discourse of development. These regions, as outlined in their respective documents, prioritise construction expansion, a comfortable urban environment creation, and jobs generation.

It is noteworthy that the compactisaton discourse, which permeates the media, does not find its reflection in the strategic planning documents of regional policy, except for the outdated Socio-Economic Development Strategy of the Murmansk Oblast', which was in action for only 3 years (Government of Murmansk Oblast', 2010). This issue is also mentioned in the strategic documents of the cities of the Komi Republic, such as Vorkuta (Vorkuta Urban District Council, 2020) and Inta (Inta Urban District Council, 2020).

Analysis of state programs showed that urban shrinkage remains primarily a local or, at most, regional issue. The case of the Komi Republic warrants special attention, as the discourse surrounding the compactisation of northern cities in this region has gained prominence in media coverage, making the problem of urban shrinkage more salient. Nonetheless, the discussion of urban shrinkage at the federal level serves as a manifestation of the imperative need to find a swift resolution to a pressing issue, rather than representing an alternative model for urban development.

This is why discourse assumes a critical role in the process of policy formulation, broadening the primary vision of a limited number of policy actors regarding the development of the Arctic and incorporating it into the goal-setting framework for all stakeholders. In light of the ongoing or impending population decline in the Arctic regions, there is a risk of misplaced priorities: a simplistic approach to 'development' based solely on extensive federal assistance in the form of financial benefits and urban environment comfortisation may lead to severe long-term consequences.

Conclusion

In this research, a qualitative content analysis of the discourse on urban shrinkage in multilevel governance in the Arctic zone of the Russian Federation has been conducted. The central theme of this work is that the discourses employed by policymakers in the Arctic significantly influence decision-making processes. Policy actors, guided by their values, beliefs, and ideological perspectives, which often entail a limited vision of reality, create problems and propose policy responses that ultimately shape urban governance. Studying discourse is crucial for comprehending how policies work: who are the key policymakers shaping the discourse, what issues they prioritise, and how they formulate solutions. Moreover, the discourse reveals the power dynamics among actors, with the most influential gaining greater visibility and influence.

The research was conducted on the Russian Arctic instance, which is a significantly urbanised territory of Russia. It faces the challenge of long-term urban depopulation, accompanied by a range of related issues. Statistical data reveal that the population within the current boundaries of the Arctic has decreased by over a third, dropping from 3.78 million to 2.41 million people. Moreover, the rate of population decline remains on the rise. In light of these trends, the question of developing Arctic regions and cities under depopulation becomes crucial. The case of the Arctic is particularly intriguing due to the abundance of policymakers who contribute to discussions on urban development.

The theoretical framework of the research is underpinned by the policy studies branch and is based on Lasswell's cyclical model of the policymaking process, developed in the 1950s, and subsequently refined by other scholars, such as Jann & Wegrich (2017). Specifically, the focus of the research is on the second stage of the policymaking process — policy formulation, which involves exploring policy alternatives, identifying problems and solutions, and elucidating how the selected solutions will address the problem at hand. The policy formulation process is further explored through the Policy Design Theory, as presented by Linder & Peters (1987), which emphasises policy design as a crucial determinant of a policy's success and effectiveness. Additionally, the work acknowledges the contributions of social constructionism scholars who have explored the discursive approach to policymaking, including Ball (1993, 2015), Fischer (2003), Yanow (1995, 2007), and Bacchi (1999, 2000, 2009).

Scholars exploring urban shrinkage from various corners of the globe have long been acutely aware of the challenge posed by the inadequate response to population decline and pro-growth policies (for instance, Wiechmann & Pallagst, 2012). This thesis endeavours to bridge the existing research gap regarding policy responses to urban shrinkage, contending that the vague

and contradictory policy formulation to the Arctic's depopulating regions and cities impedes the implementation of adaptive responses.

This thesis is based on an analysis of news media publications featuring statements by Arctic officials from 2014 to 2019 and 2020 to 2024. It delves into the key trends and conducts an examination of a sample comprising 106 publications, employing qualitative content analysis (QCA) methodology. The analysis uncovers a notable increase in references to population decline in the recent period. Moreover, it highlights the inherent contradictions in the discourse, where officials attribute macroeconomic factors as the root cause of the challenges associated with population decline, yet they propose more readily implementable measures in the housing sector as solutions.

The final discourses (themes) of QCA: 'Stop shrinkage via providing financial benefits', 'Stop shrinkage via comfortisation', and 'Mitigate urban shrinkage problems through compactisation'. An analysis of actor mentions reveals a significant role played by a narrow circle of policymakers in shaping these discourses, including federal officials and governors of the regions vying for the status of flagship Arctic regions, such as the Murmansk Oblast' and Arkhangel'sk Oblast'. Concurrently with discourses on halting shrinkage, there is a growing discourse on compactisation championed by the Komi Republic and its cities of Vorkuta and Inta. The compactisation approach is not considered as a replacement for growth strategy, but rather as an urgent solution to address housing sector acute challenges.

The research is underpinned by a comprehensive analysis of both current and previous strategic planning documents across all territorial levels, encompassing federal, regional, and local jurisdictions. A total of 114 documents, including strategies for socio-economic development and state programs at both the federal and regional scales, were examined. The increasing frequency of mentions of terms associated with population decline underscores the heightened media attention devoted to this issue in recent years.

The final discourses (themes): 'Stop the population decline at all costs', 'Comfortisation turn for urban growth', and 'New jobs as an intangible remedy against shrinkage'. The analysis reveals that federal objectives are consistently reflected in regional and local planning, irrespective of the specific context in which a region or city operates. However, it is crucial to acknowledge that the measures proposed in regional state programs may be sensitive to the specifics of shrinkage.

The role of discourse on urban shrinkage in the multilevel governance of the Arctic therefore, is that it serves as a means of translating the vision of key policy actors — federal authorities, governors of Murmansk Oblast' and Arkhangel'sk Oblast', who advocate for growth and comfort — into a comprehensive Arctic perspective. This process is reflected in strategic

goal-setting and unifies policies across all Arctic regions, including the Komi Republic engaged in urban shrinkage practices. This alignment of efforts underscores the inconsistency of policy formulation, where goal-setting may not meet the expected results. Moreover, it raises questions about the realisability of urban policies in the Arctic.

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Appendix 1

Quantitative assessment of shrinkage is conventional and used to conceptualise shrinking regions and cities and to simplify the analysis.

$$CAGR = \left[\left(\frac{V\,final}{V\,begin}\right)^{\frac{1}{t}} - 1\right] * 100\% \ ,$$

CAGR — compound annual growth rate
V final — final value
V begin — beginning value
t — time in years

Appendix 2

Table 1. CAGR for 1989-2021 and 2010-2021 intercensal periods

№	City name	Population of the municipality, 2021	CAGR (1989-2021), %	CAGR (2010-2021), %	Current shrinkage status
1	Apatity	49 663	-1,95	-1,66	Severely shrinking
2	Arkhangel'sk	306 021	-0,95	-1,36	Severely shrinking
3	Belomorsk	13 076	-2,15	-3,39	Severely shrinking
4	Beloyarskiy	28 897	1,07	-0,35	Moderately shrinking
5	Bilibino	7418	-2,82	-0,53	Moderately shrinking
6	Dikson	306	-8,02	-6,95	Severely shrinking
7	Dudinka	30 154	-1,87	-1,2	Severely shrinking
8	Igarka	12 808	-1,27	-3,39	Severely shrinking
9	Inta	21 857	-3,41	-4,23	Severely shrinking
10	Kandalaksha	40 981	-1,55	-1,71	Severely shrinking
11	Kem'	13 138	-1,72	-2,7	Severely shrinking
12	Kirovsk	26 852	-1,5	-1,29	Severely shrinking
13	Kola	34 112	-0,99	-2,42	Severely shrinking
14	Kostomuksha	26 650	-0,41	-0,76	Severely shrinking
15	Kovdor	17 182	-1,78	-1,93	Severely shrinking
16	Mezen'	7 466	-0,92	-2,91	Severely shrinking
17	Mochegorsk	42 154	-1,51	-1,17	Severely shrinking
18	Muravlenko	29 233	0,74	-1,2	Severely shrinking
19	Murmansk	270 384	-1,7	-1,16	Severely shrinking
20	Novodvinsk	33 294	-1,27	-1,79	Severely shrinking
21	Noyabr'sk	100 188	0,48	-0,9	Severely shrinking
22	Olenegorsk	28 159	-0,73	-0,58	Moderately shrinking
23	Onega	24 922	-0,95	-3,13	Severely shrinking
24	Pevek	4 776	-5,03	-1,04	Severely shrinking
25	Polyarnyye Zori	15 818	-0,64	-0,99	Severely shrinking

26	Segezha	32 371	-1,45	-2,17	Severely shrinking
27	Severodvinsk	157 925	-1,41	-1,83	Severely shrinking
28	Severomorsk	50 796	-1,86	-2,53	Severely shrinking
29	Srednekolymsk	6 805	1,31	-1,34	Severely shrinking
30	Tarko-Sale	36 569	0,7	-3,03	Severely shrinking
31	Tiksi	7 706	-1,28	-1,45	Severely shrinking
32	Usinsk	36 913	-1,37	-2,22	Severely shrinking
33	Vorkuta	68 425	-3,51	-3,02	Severely shrinking
34	Zapolyarnyy	32 053	-1,3	-1,75	Severely shrinking

Source: author's work based on population census data for 2010, 2021, and Eva et al. (2021)

Appendix 3

Table 1. Number of media publications within the 2014-2019 period

	2014	2015	2016	2017	2018	2019	**Total**
Governor of the ARK	50	16	58	52	20	78	**274**
Head of the SAH	35	60	58	64	28	2	**247**
Governor of the MUR	75	38	1	22	63	43	**242**
Head of the KOM	0	23	38	40	31	73	**205**
Head of the KAR	37	30	67	20	1	6	**161**
Governor of the KHM	4	1	0	8	1	60	**74**
Governor of the YAN	3	14	14	9	2	10	**52**
Governor of the KYA	5	0	0	2	1	42	**50**
Governor of the CHU	3	0	1	13	6	4	**27**
Governor of the NEN	1	0	0	2	0	0	**3**

Table 2. Number of media publications within the 2020-2024 period

	2020	2021	2022	2023	2024	**Total**
Governor of the MUR	123	313	38	163	303	**940**
Governor of the ARK	243	33	14	251	20	**561**
Head of the SAH	172	140	34	127	73	**546**
Head of the KOM	103	71	11	32	66	**283**
Head of the KAR	74	18	53	16	24	**185**
Governor of the KYA	0	0	0	36	13	**49**
Governor of the NEN	8	1	0	7	1	**17**
Governor of the YAN	1	1	2	2	3	**9**

Governor of the KHM	0	0	0	0	7	**7**
Governor of the CHU	0	1	0	3	2	**6**

Appendix 4

Table 1. Frequency of shrinkage words in Socio-Economic Development Strategies

	Outflow		Decline		Decrease		Depopulation		Migration loss		Natural loss		Fall		Shrinkage	
Documents (outdated / in action)	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓
1. Federal Socio-Economic Development Strategies																
Government of the Russian Federation (2019, 2024)	4	8	0	6	3	5	0	0	4	0	0	5	0	0	0	0
Government of the Russian Federation (2009)	7	-	2	-	9	-	1	-	4	-	2	-	0	-	0	-
Government of the Russian Federation (2011a)	11	-	6	-	8	-	1	-	4	-	2	-	3	-	0	-
Government of the Russian Federation (2011b)	3	-	6	-	6	-	1	-	3	-	5	-	4	-	0	-
Government of the Russian Federation (2023)	-	1	-	1	-	1	-	0	-	1	-	1	-	0	-	0
President of the Russian Federation (2008, 2020a)	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0

	Outflow		Decline		Decrease		Depopulation		Migration loss		Natural loss		Fall		Shrinkage	
Documents (outdated / in action)	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓
President of the Russian Federation (2013, 2020b)	1	2	0	2	0	0	0	0	0	2	0	1	0	0	0	0
Government of the Russian Federation (2022)	-	1	-	2	-	0	-	0	-	0	-	0	-	2	-	0
2. Regional Socio-Economic Development Strategies																
Government of the Komi Republic (2006, 2019a)	4	13	2	9	7	4	0	1	20	16	4	13	3	5	0	1
Government of the Murmansk Oblast' (2010, 2013b)	11	5	7	1	29	4	0	0	8	5	3	2	0	3	9	0
Government of the Karelia Republic (2007, 2018)	4	10	1	7	8	16	0	3	6	13	2	4	0	0	0	1
The Head of the Sakha Republic (Yakutia) (2018)	-	5	-	1	-	9	-	0	-	5	-	0	-	0	-	1
The Head of the Sakha Republic (Yakutia) (2020)	-	6	-	2	-	7	-	0	-	5	-	5	-	0	-	0
Administration of the Arckhangel'sk Oblast' (2008),	7	5	0	1	4	13	0	0	6	5	0	1	2	1	0	0

	Outflow		Decline		Decrease		Depopulation		Migration loss		Natural loss		Fall		Shrinkage	
Documents (outdated / in action)	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓
Governor of the Archangel'sk Oblast' (2019)																
Government of the Khanty-Mansi Autonomous Okrug - Yugra (2013, 2022)	0	12	0	3	1	8	0	0	0	3	0	6	0	2	0	0
Legislative Assembly of the Yamal-Nenets Autonomous Okrug (2011, 2021)	0	37	0	0	0	8	0	0	0	29	0	2	0	7	0	0
Assembly of the Deputies of the Nenets Autonomous Okrug (2019)	-	11	-	1	-	5	-	2	-	8	-	0	-	6	-	0
Government of the Chukotka Autonomous Okrug (2014)	-	5	-	0	-	4	-	0	-	5	-	0	-	1	-	0
Government of the Krasnoyarsk Kray (2013a, 2018)	4	8	5	3	13	7	1	3	5	4	3	4	1	2	1	0
3. Municipal Socio-Economic Development Strategies																
Council of Deputies of the Apatity (2020)	-	15	-	30	-	8	-	0	-	13	-	11	-	2	-	0

	Outflow		Decline		Decrease		Depopulation		Migration loss		Natural loss		Fall		Shrinkage	
Documents (outdated / in action)	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓
Arkhangel'sk City Mayor (2008), Arkhangel'sk City Duma (2022)	1	5	7	11	5	14	0	0	2	4	1	2	1	3	1	0
Administration of the Belomorsk Municipal District (2024)	-	2	-	4	-	2	-	0	-	0	-	2	-	0	-	0
Beloyarskiy Municipal District Duma (2014, 2023)	8	5	8	6	2	5	0	0	5	5	0	0	0	1	0	0
Bilibino Municipal District Strategy Project (2025)[6]	-	0	-	1	-	0	-	0	-	0	-	1	-	0	-	0
Taymyr Dolgano-Nenets Municipal District Council of Deputies (2019)	-	3	-	2	-	3	-	2	-	1	-	0	-	1	-	0
Turukhanskiy Municipal District Strategy Project (2020)[7]	-	8	-	11	-	4	-	0	-	2	-	0	-	2	-	0
Inta Urban District	9	18	15	19	10	5	0	0	6	5	2	1	1	2	0	0

[6] There is no information about the Bilibino Municipal District Socio-Economic Development Strategy approval
[7] There is no information about the Turukhanskiy Municipal District Socio-Economic Development Strategy approval

	Outflow		Decline		Decrease		Depopulation		Migration loss		Natural loss		Fall		Shrinkage	
Documents (outdated / in action)	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓
Council (2014, 2020)																
Kandalaksha Municipal District Council of Deputies (2018).	-	8	-	23	-	11	-	0	-	10	-	9	-	0	-	0
Kem' Municipal District Council (2016, 2022).	0	19	4	14	2	6	0	0	0	9	2	10	0	0	0	0
Kirovsk Council of Deputies (2011), Administration of the Kirovsk Municipal District (2024)	2	2	10	20	2	4	2	0	8	11	3	7	0	0	0	0
Kola Municipal District Council of Deputies (2015)	6	-	6	-	7	-	0	-	4	-	3	-	2	-	0	-
Kostomuksha Urban District Council (2008, 2020).	7	5	5	7	2	7	0	0	1	4	0	2	1	2	0	0
Mezen' Municipal District Assembly of Deputies (2020)	-	11	-	14	-	6	-	0	-	5	-	5	-	1	-	0
Administration of the Monchegorsk Municipal District (2020, 2023)	1	2	19	18	2	0	0	0	8	4	10	4	1	0	0	0

	Outflow		Decline		Decrease		Depopulation		Migration loss		Natural loss		Fall		Shrinkage	
Documents (outdated / in action)	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓
Administration of the Muravlenko City (2011)	5	-	4	-	5	-	0	-	4	-	2	-	4	-	1	-
Murmansk City Council of Deputies (2012, 2018)	11	27	10	18	22	26	0	0	5	19	0	2	4	4	0	0
Noyabr'sk City Duma (2014)	20	-	16	-	9	-	0	-	6	-	0	-	7	-	1	-
Administration of the Olenegorsk City (2019, 2022)	2	2	10	12	3	2	0	0	2	2	2	4	0	0	0	0
Onega Municipal District Assembly of Deputies (2017)	-	7	-	15	-	5	-	0	-	8	-	0	-	0	-	0
Administration of the Pevek Urban District (2020, 2024)	1	3	1	4	0	1	0	0	2	4	0	0	3	1	0	0
Administration of the Polyarnyye Zori City (2019, 2024)	2	0	10	11	1	4	0	0	4	2	0	3	0	0	0	0
Segezha Municipal District Council (2015, 2021)	22	13	10	20	12	5	0	0	5	8	2	7	1	2	0	0
Severodvinsk City District Council of	-	7	-	20	-	13	-	0	-	10	-	4	-	0	-	0

	Outflow		Decline		Decrease		Depopulation		Migration loss		Natural loss		Fall		Shrinkage	
Documents (outdated / in action)	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓
Deputies (2019)																
Severomorsk Closed Administrative-Territorial Entity Council of Deputies (2019)	-	3	-	2	-	3	-	0	-	1	-	0	-	0	-	0
Srednekolymsk Municipal District Council of Deputies (2018)[8]	-	1	-	5	-	3	-	1	-	2	-	0	-	0	-	0
Purovskiy Municipal District Duma (2011, 2018)	5	1	16	1	4	0	0	0	5	0	0	0	0	0	0	0
Bulunskiy Municipal District Program Project (2010)[9], Bulunskiy Municipal District Council of Deputies (2018)	1	10	1	7	1	9	0	0	1	9	0	0	0	0	0	0
Usinsk Urban District Council (2014, 2020)	6	4	7	8	5	5	0	0	5	5	2	1	0	1	0	0
Vorkuta Urban District Council	2	34	11	57	2	7	0	1	1	14	1	2	2	3	0	4

[8] This is a project of Srednekolymsk Municipal District Council of Deputies By-law
[9] There is no information about the Bulunskiy Municipal District Socio-Economic Development Program approval

	Outflow		Decline		Decrease		Depopulation		Migration loss		Natural loss		Fall		Shrinkage	
Documents (outdated / in action)	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓	✗	✓
(2010, 2020).																

Sourse: author's work

Appendix 5

Table 1. Housing construction indicators

	Housing, sq. m	Housing projection	CAGR, %	Housing, sq. m	Housing projection	CAGR, %
KOM	122	264	10,13	234,5	258,5	0,89
MUR	24,3	40	7,38	44,8	85	7,38
KAR	-	-	-	312	380	2,86
SAH	520	608	3,99	705,1	-	-
ARK	-	-	-	440	-	-
KHM	898	1154	4,27	955	1582	7,48
YAN	-	-	-	158	192	4,99
NEN	-	-	-	20,8	37	8,58
CHU	1	2	10,41	3,52	-	-
KYA	-	-	-	1631	1800	1,42

Sourse: author's work